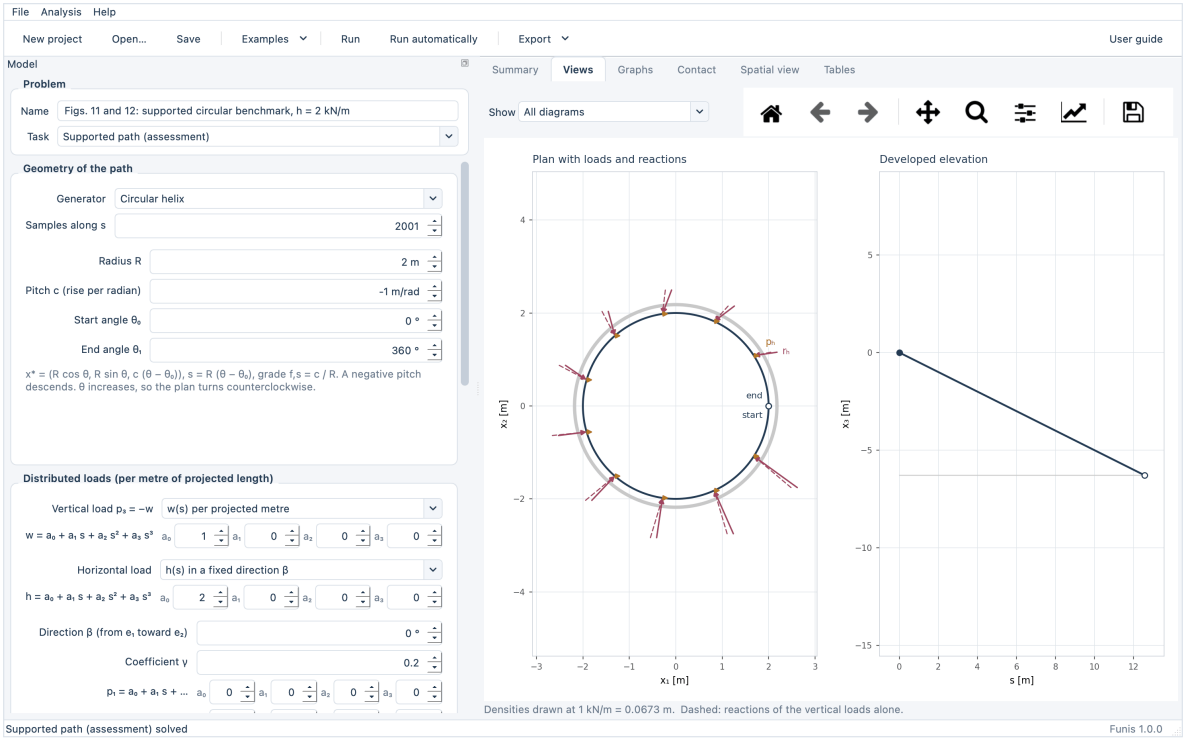
Funis

User Manual

Equilibrium and form-finding of spatial compression paths:
the analytical Linear Arch Static Analysis (LASA), plane funiculars
and the thrust line assessment of arches



Version 1.0.0

8 November 2026

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DOI 10.5281/zenodo.24040974 • https://github.com/colivieri89/Funis

Funis 1.0.0 User Manual
8 November 2026

This manual describes version 1.0.0 of Funis, the desktop application for Windows and macOS of the analytical Linear Arch Static Analysis of spatial compression paths, with the plane funicular and the thrust line assessment of masonry arches. It is rebuilt for every release of the program, and the edition matching each version is kept in the repository as `docs/Funis_User_Manual.pdf`.

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Repository. https://github.com/colivieri89/Funis

Archive. Every release is archived on Zenodo. The concept DOI 10.5281/zenodo.24040974 refers to all versions and resolves to the latest one.

How to cite. Carlo Olivieri, *Funis: equilibrium and form-finding of spatial compression paths*, version 1.0.0, software, doi:10.5281/zenodo.24040974. The file `CITATION.cff` of the repository gives the same data in machine-readable form.

Licence. Copyright © 2026 Carlo Olivieri. Funis is released under the MIT License, given in the file `LICENSE` of the repository. This manual is part of the repository and is distributed under the same licence. The installers also contain third-party libraries under their own licences, among them Qt for Python (PySide6, LGPL-3.0), NumPy, SciPy and Matplotlib.

Disclaimer. The software is provided "as is", without warranty of any kind, as stated in the licence. Funis computes equilibrium states under the hypotheses described in this manual. Whether those hypotheses represent a given structure, and every decision taken on the basis of the results, remain the responsibility of the user.

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1 Introduction

1.1 What Funis does

Funis computes the equilibrium of continuous compression paths, that is, of lines that carry their loads by an axial force alone. In space it follows the analytical Linear Arch Static Analysis (LASA) of C. Olivieri [1]: a path is either prescribed, and the program finds the axial force and the lateral reactions that keep it in equilibrium, or unknown, and the program finds the curve from the loads and the force at its start. In the vertical plane it draws the funicular line of given loads, the catenary and the parabola among them, and assesses masonry arches of finite thickness by their thrust lines.

The program is a desktop application for Windows and macOS. It needs no other software: the installers contain everything, Python included. A model is entered in a panel of input sections, solved in a fraction of a second, and read in a summary, in drawings to scale, in graphs along the line, in contact diagrams, in a rotatable spatial view and in tables. Every drawing and table can be exported, and a self-contained HTML report collects the whole analysis.

1.2 The six tasks

Table 1.1 lists the tasks. The first four act on spatial paths and follow the article. The last two act in the vertical plane.

Table 1.1. Tasks of Funis.

Task	Question answered	Chapter
Supported path (assessment)	A path of given shape leans on a lateral support, such as the wall of a stair well. Which axial force and which reactions keep it in equilibrium, and can the support supply them within a friction cone?	6
Path without reactions	Is a path of given shape an axial equilibrium of given loads with no reaction at all, and which loads would make it one?	7
Unknown geometry (form-finding)	Which curve carries given loads in pure compression from a given start point and initial force, or between two given supports?	8
Node under a concentrated force	How does a concentrated force change the force and the direction of the line at a node?	9
Plane funicular (catenary)	Which plane line carries given vertical and horizontal loads in compression between two supports?	10
Arch assessment (thrust line)	Does a thrust line fit within a masonry arch, which thrusts are admissible, and how far is the arch from collapse?	11

1.3 Scope and relation to the article

The equations of the spatial tasks are those of the article [1], and the cases of the article are available in the **Examples** menu, where figure and section numbers refer to it. The plane funicular and the thrust line assessment of arches are tasks of the application. They rest on classical statics of plane funiculars and on Heyman's safe theorem for masonry arches [3, 4], and they are not results of the article. Chapter 16 lists the hypotheses of every task.

1.4 About this manual

Chapters 2 to 5 cover installation, the interface and the conventions common to every task. Chapters 6 to 11 describe the tasks one by one: the input sections with their fields, the method, the results and a worked example taken from the **Examples** menu. Chapters 12 to 16 cover the checks of every result, the exports, the catalogue of examples, the messages of the program and the limitations. Appendix A collects the keyboard shortcuts, the command line and the notation.

Interface elements are set in a sans-serif face: menu commands as **File › Export › Report (HTML)…**, fields and buttons as Preload T = −S(0), keys as F5. File names are set as `project.funis`. The screenshots were taken on macOS. On Windows the window has the same layout, with the fonts of the system.

The values printed in the worked examples and the screenshots are produced by the program itself when this manual is built, so they are those that version 1.0.0 gives.

2 Installation

2.1 Requirements

The installers of every release are published on the release page of the repository, https://github.com/colivieri89/Funis/releases/latest:

- `Funis-<version>-Windows-Setup.exe` for 64-bit Windows 10 or 11,
- `Funis-<version>-macOS-arm64.pkg` and `Funis-<version>-macOS-arm64.dmg` for Macs with Apple silicon.

On any other system, including Macs with an Intel processor and Linux, Funis runs from its source code with Python 3.9 or later (Section 2.4). No licence key, account or network connection is needed.

2.2 Windows

1. Download `Funis-<version>-Windows-Setup.exe` and run it.
2. The installer is not signed, so Windows SmartScreen may show the message *Windows protected your PC*. Choose More info and then Run anyway.
3. Follow the wizard. Funis is installed in `Program Files\Funis`, with an entry in the Start menu, an optional icon on the desktop and the association of the `.funis` files, so that a double click on a project opens it in Funis.

To uninstall Funis use Uninstall Funis in the Start menu, or Settings › Apps.

2.3 macOS

1. Download `Funis-<version>-macOS-arm64.pkg` and open it. The installer copies `Funis.app` into `Applications`. Alternatively, open the `.dmg` file and drag `Funis.app` into `Applications`.
2. The application is not notarised. At the first launch, macOS may refuse to open it. Open it with a right click on `Funis.app` and Open, or allow it in System Settings › Privacy & Security with Open Anyway. Later launches need no confirmation.

To uninstall Funis move `Funis.app` from `Applications` to the Bin. On macOS, open project files with **File › Open…**.

2.4 Running from the source code

With Python 3.9 or later, in the folder of the repository:

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
.venv/bin/python run_funis.py          # or: .venv/bin/python -m funis
```

On Windows use `.venv\Scripts\python` instead of `.venv/bin/python`. The dependencies are NumPy, SciPy, Matplotlib, PySide6, openpyxl, ezdxf and Pillow. The test suite of the repository runs with

```
.venv/bin/python -m pip install pytest
.venv/bin/python -m pytest -q
```

2.5 Checking an installation

The option `--selftest` opens the window without showing it, solves every example and exits with code 0 when all of them solve:

```
"C:\Program Files\Funis\Funis.exe" --selftest C:\Temp\funis.png      (Windows)
/Applications/Funis.app/Contents/MacOS/Funis --selftest ~/funis.png    (macOS)
```

The optional image receives a screenshot of the window, and a report with the same name and the extension `.log` is written next to it, for instance *selftest: 33 examples, 0 failures*. The build scripts of the installers run the same test on the packaged program.

2.6 Settings

Funis remembers the size and position of the window, the state of **Run automatically** and the last folder used in a file dialog. They are stored with the user's preferences by Qt and contain no project data.

3 Getting started

3.1 The main window

At start, Funis opens the circular benchmark of the article (Figs. 11 and 12) and solves it. Fig. 3.1 shows the window. The Model panel on the left holds the input, the results fill the rest of the window, and every change of the input is solved again after a short pause.

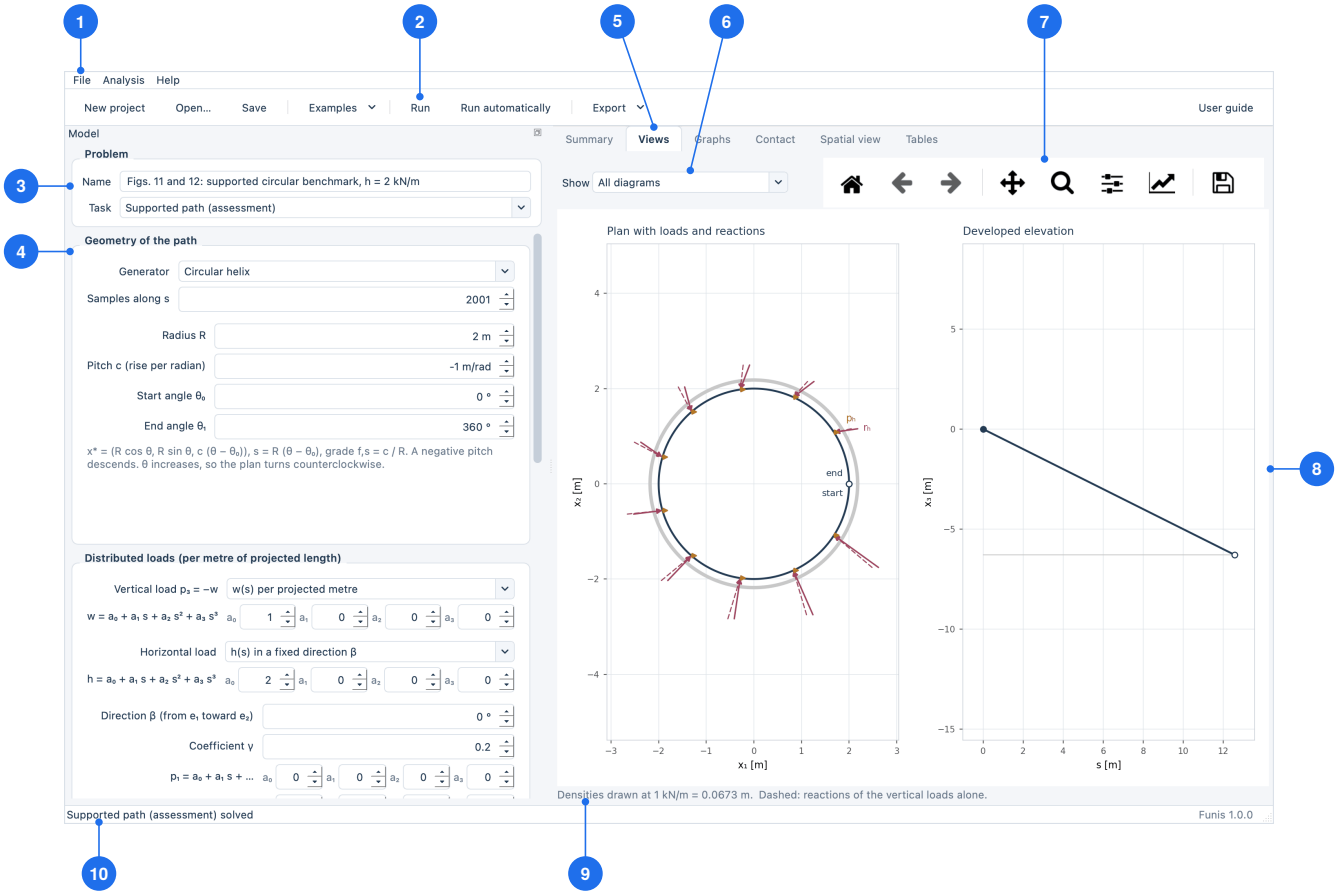


Figure 3.1. The main window with the circular benchmark of the article, tab Views.

1 **Menu bar.** **File** (projects, examples, exports), **Analysis** (run) and **Help** (user guide, about).

2 **Tool bar.** The most frequent commands: New project, Open…, Save, Examples, Run, Run automatically, Export and User guide.

3 **Problem.** The name of the project and the Task. The task decides which input sections appear below.

4 **Input sections.** The data of the task, in sections that scroll vertically: geometry, loads, point actions, support, contact, output and so on (Section 4.2).

5 **Result tabs.** Summary, Views, Graphs, Contact (Admissible thrusts for an arch), Spatial view and Tables. A tab without content for the task is disabled.

6 **Selector.** All diagrams of the tab, or one of them enlarged.

7 **Navigation.** The Matplotlib tool bar of the drawing: reset, back, forward, pan, zoom, margins, axis options and save.

8 **Drawings.** Views to scale, with the labels placed clear of the lines and of each other.

9 **Notes.** The scales of the arrows and the meaning of dashed lines in the current drawings.

10 **Status bar.** The outcome of the last run, for instance *Supported path (assessment) solved, 1 warning*, the messages of the exports and the version of the program.

3.2 A first analysis

The following steps use the benchmark that opens at start. They take a few minutes and touch every part of the interface.

1. Read the summary. Open the Summary tab. The thrust S grows from -8 kN at the start to about -33 kN at the end, the line is in compression everywhere, and the contact margin is positive: the wall can supply the required reactions with the friction coefficient $\mu = 0.6$.
2. Change a load. In Distributed loads, raise the coefficient a_0 of the horizontal load h from 2 to 4 kN/m. After a short pause the analysis runs again. The summary now shows a warning in colour: the required reaction leaves the contact cone. The line is still in compression, but the assumed wall cannot supply the reaction (Section 6.4).
3. Find the admissible preloads. The summary line Admissible preloads at λ = 1 gives the interval of the preload T for which the contact cone is satisfied with the new load, if any. Open the Contact tab and select Admissible preload and load multiplier in the selector: the shaded region contains every admissible pair (T, λ), and the marker shows the model.
4. Look at the line in space. Open Spatial view and drag the drawing to rotate it.
5. Export a report. Choose **File › Export › Report (HTML)…**, and open the file in a web browser. It holds the summary, every drawing, every table and the input.
6. Save the project. Choose **File › Save** (Ctrl+S, ⌘S on macOS). The project is written as a `.funis` file that **File › Open…** reads back.

3.3 Projects

A project holds the whole input of a task: its name, the task and the data of every section. It does not hold results, which are computed again when the project is opened.

New project (Ctrl+N) clears the input to the defaults of the program.
Open… (Ctrl+O) reads a `.funis` file, or a `.json` file with the same content.
Save (Ctrl+S) and **Save as…** (Ctrl+Shift+S) write the project. The title bar shows the file name, with an asterisk when there are unsaved changes.
Examples load one of the 33 examples of Chapter 14. An example is an ordinary project: modify it and save it under a new name.

Before a command replaces a project with unsaved changes, and before the window closes, Funis asks whether to discard them. On macOS the shortcuts use ⌘ in place of Ctrl.

4 Working with the interface

4.1 Menus and tool bar

Table 4.1. Commands of the menus. The tool bar repeats the most frequent ones.

Command	Shortcut	Action
File › New project	Ctrl+N	Clear the input to the defaults.
File › Open…	Ctrl+O	Open a project (`.funis` or `.json`).
File › Examples		Load an example (Chapter 14).
File › Save	Ctrl+S	Save the project.
File › Save as…	Ctrl+Shift+S	Save the project under a new name.
File › Export › Drawings (PDF, SVG, PNG)…		Write the selected drawings (Section 13.1).
File › Export › Drawings as DXF…		Write the views to scale and the spatial path to one DXF file.
File › Export › Tables as CSV…		Write every table to one CSV file.
File › Export › Tables as Excel…		Write every table to a sheet of one workbook.
File › Export › Results as JSON…		Write the input, the summary, the tables and the messages.
File › Export › Report (HTML)…		Write a self-contained report.
File › Quit	Alt+F4	Close Funis (Alt+F4 is the window command of Windows).
Analysis › Run	F5	Solve the model now.
Analysis › Run automatically		Solve the model after every change (on by default).
Help › User guide	F1	The short guide of the program, with the equations of every task.
Help › About Funis		Version, licence and reference.

The shortcuts are those of Windows. On macOS ⌘ replaces Ctrl, ⌘? opens the user guide, and **Quit** (⌘Q) is in the application menu (Section A.1). The Examples and Export buttons of the tool bar open the same lists as the menus.

4.2 The Model panel

The panel opens with the Problem box (Fig. 4.1): a free Name, printed as the title of the summary and of the report, and the Task. Below it, the sections of the task appear in a fixed order (Table 4.2), and the panel scrolls when they do not fit. The panel can be detached from the window with the button at its top right and docked again by a double click on its title.

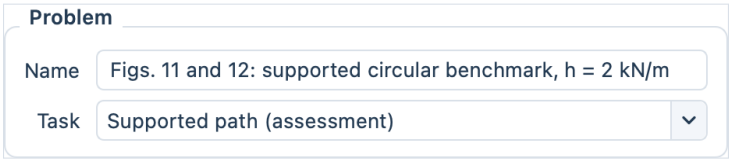


Figure 4.1. The Problem box.

Table 4.2. Input sections of every task, in the order of the panel.

Task	Sections
Supported path (assessment)	Geometry of the path, Distributed loads, Point actions, Support of the line, Contact with the wall, Output
Path without reactions	Geometry of the path, Path without reactions, Distributed loads and Point actions (only to check given loads), Output
Unknown geometry (form-finding)	Free curve, Distributed loads, Point actions, Output
Node under a concentrated force	Node
Plane funicular (catenary)	Plane funicular, Loads on the line, Point loads, Output
Arch assessment (thrust line)	Arch, Fill and surcharge, Thrust line assessment, Point loads, Output

4.3 Entering values

Numbers. A numeric field shows its unit after the value. Type the value, or use the arrows and the mouse wheel. Lengths are in metres, forces in kilonewtons, angles in degrees.

Choices. A drop-down list selects a variant, for instance the generator of the geometry or the kind of horizontal load. The fields that the variant does not use are disabled or hidden.

Polynomials. A load or a thrust given as a polynomial has the four coefficients a_0 to a_3 of $a_0 + a_1 s + a_2 s^2 + a_3 s^3$, with s in metres. A constant is entered in a_0 alone.

Vectors. A vector has one field per component, for instance C_1, C_2, C_3 of an initial force.

Tables. Points, load tables and point actions are entered in tables. Add row appends a copy of the last row, Remove deletes the selected rows (or the last one), and Fill column copies the value of the selected cell to its whole column. A cell accepts a decimal point or a decimal comma. A cell that is not a number stops the run with a message naming its row and column. A table of loads is interpolated linearly between its rows and held constant beyond its ends.

4.4 Running the analysis

With **Analysis › Run automatically** on, every change is solved 0.35 s after the last keystroke, so the results follow the input while it is typed. With the option off, or for a large number of samples, press F5 or Run. The status bar reports the outcome:

- *<task> solved*, possibly with the number of warnings, which the Summary tab lists in colour above the results,
- *Input error*, when a field cannot be read, or *The task could not be solved*, with the reason, when the data do not admit a solution of the task. The Summary tab shows the message (Chapter 15).

4.5 Reading the results

Summary. The name, the task, the notes of an example, the warnings and a table of the main results (Fig. 4.2). The values can be selected and copied.

Views. Plans, elevations and other drawings to scale. Distributed loads and reactions are drawn as arrows at the scale given in the notes below the drawing.

Graphs. Quantities along the line: forces, reactions, loads, residuals, curvature and torsion, the eccentricity of an arch.

Contact. For a supported path with the contact check on: the contact margin, the contact cone, the admissible region of preload and load multiplier, the admissible horizontal loads and the minimum preload. For an arch the tab is named Admissible thrusts and shows the range of the thrust against the multiplier of the variable loads.

Spatial view. The line in space with its loads and reactions. Drag to rotate, drag with the right button to zoom.

Tables. The values at evenly spaced stations, the balance checks, the summary and the tables of the task (point actions, endpoint correction, joints, eccentricities, hinges, loads).

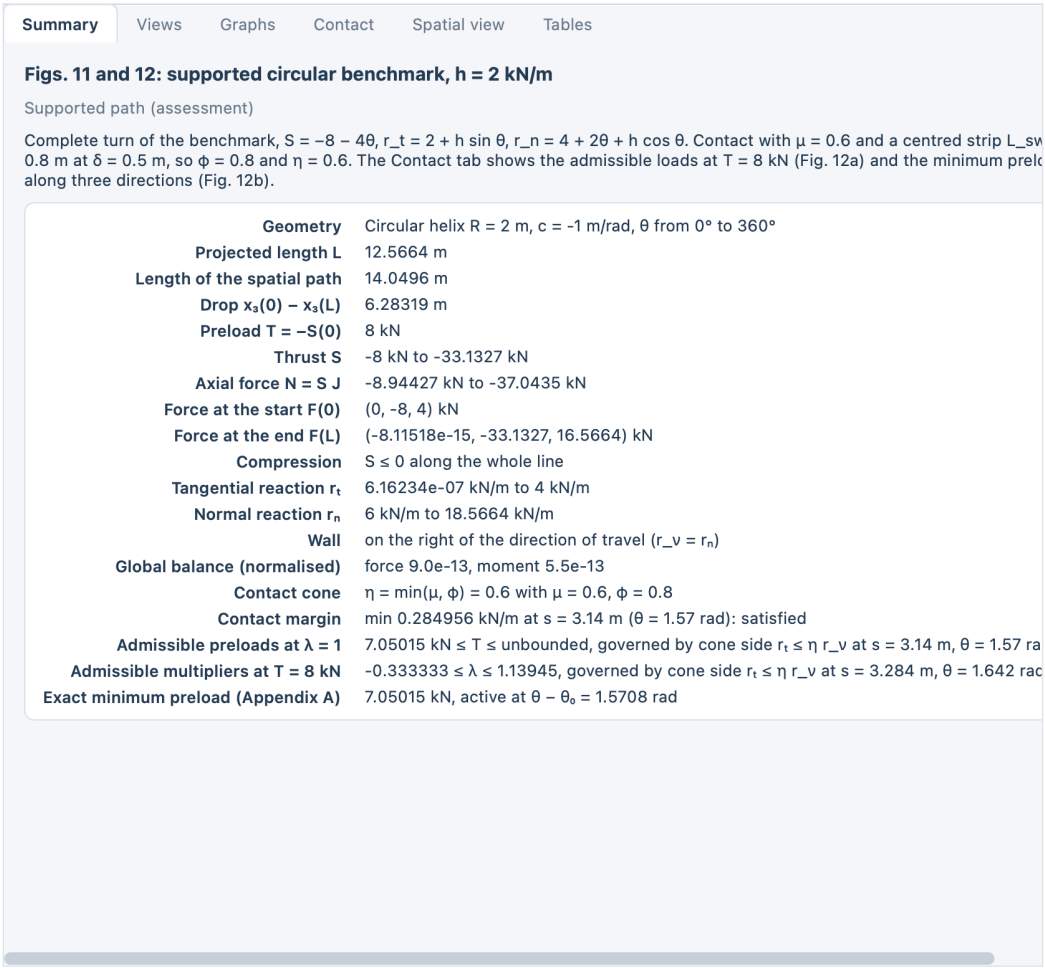
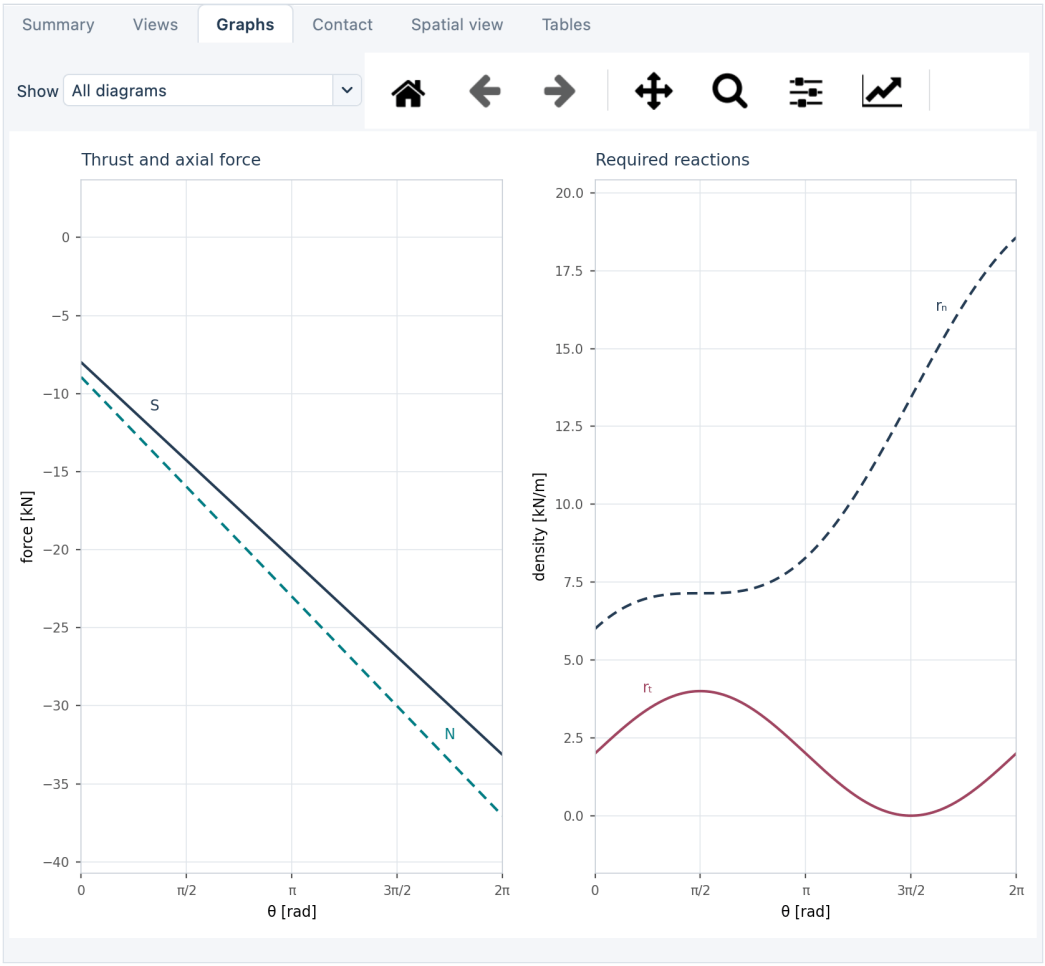
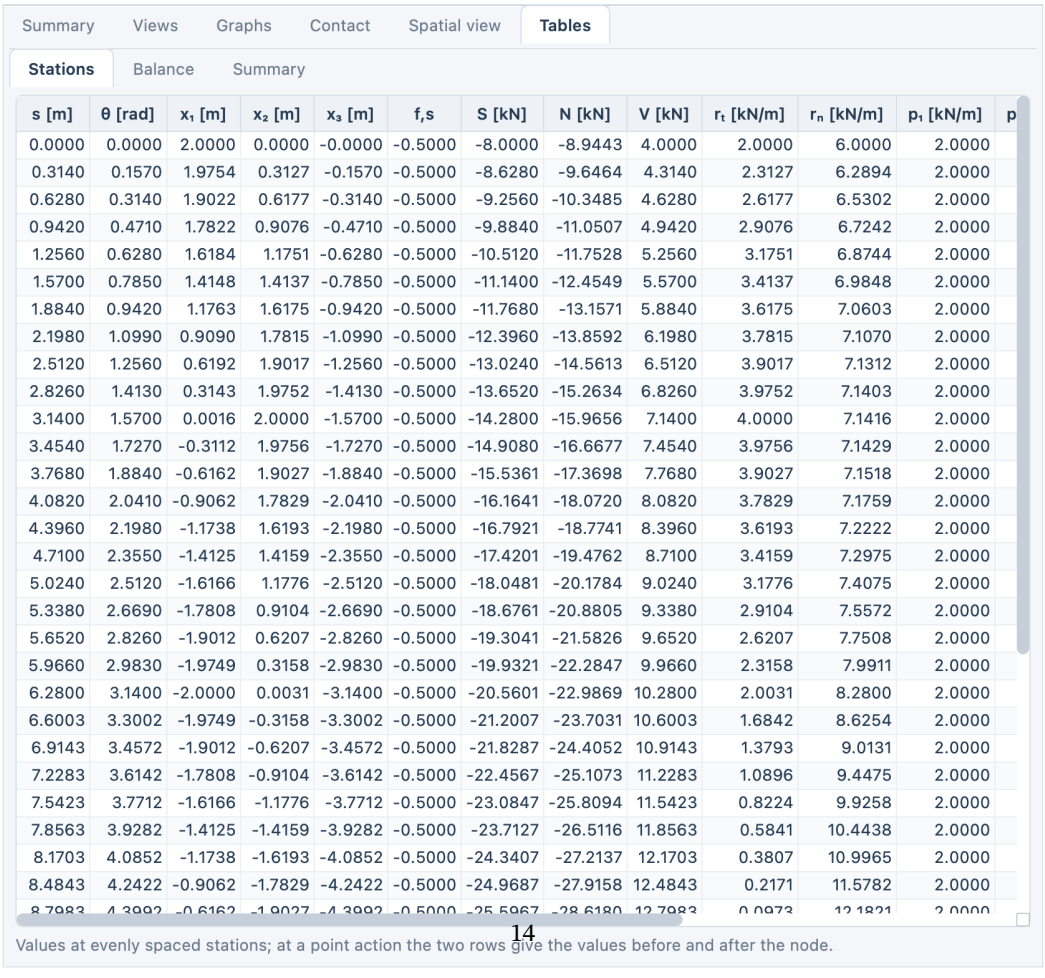


Figure 4.2. The Summary tab of the benchmark.

In Views, Graphs and Contact, the selector Show displays All diagrams of the tab in a grid, or one of them enlarged. The tool bar at the right of the selector is that of Matplotlib: Home restores the full view, the arrows step through previous views, the cross pans, the magnifier zooms to a rectangle, and the disk saves the visible figure as an image. Labels are placed again at every redraw, clear of the lines, arrows and other labels, so they stay readable at any size of the window.





s [m]	θ [rad]	x_1 [m]	x_2 [m]	x_3 [m]	f,s	S [kN]	N [kN]	V [kN]	r_t [kN/m]	r_n [kN/m]	p_1 [kN/m]	p
0.0000	0.0000	2.0000	0.0000	-0.0000	-0.5000	-8.0000	-8.9443	4.0000	2.0000	6.0000	2.0000	
0.3140	0.1570	1.9754	0.3127	-0.1570	-0.5000	-8.6280	-9.6464	4.3140	2.3127	6.2894	2.0000	
0.6280	0.3140	1.9022	0.6177	-0.3140	-0.5000	-9.2560	-10.3485	4.6280	2.6177	6.5302	2.0000	
0.9420	0.4710	1.7822	0.9076	-0.4710	-0.5000	-9.8840	-11.0507	4.9420	2.9076	6.7242	2.0000	
1.2560	0.6280	1.6184	1.1751	-0.6280	-0.5000	-10.5120	-11.7528	5.2560	3.1751	6.8744	2.0000	
1.5700	0.7850	1.4148	1.4137	-0.7850	-0.5000	-11.1400	-12.4549	5.5700	3.4137	6.9848	2.0000	
1.8840	0.9420	1.1763	1.6175	-0.9420	-0.5000	-11.7680	-13.1571	5.8840	3.6175	7.0603	2.0000	
2.1980	1.0990	0.9090	1.7815	-1.0990	-0.5000	-12.3960	-13.8592	6.1980	3.7815	7.1070	2.0000	
2.5120	1.2560	0.6192	1.9017	-1.2560	-0.5000	-13.0240	-14.5613	6.5120	3.9017	7.1312	2.0000	
2.8260	1.4130	0.3143	1.9752	-1.4130	-0.5000	-13.6520	-15.2634	6.8260	3.9752	7.1403	2.0000	
3.1400	1.5700	0.0016	2.0000	-1.5700	-0.5000	-14.2800	-15.9656	7.1400	4.0000	7.1416	2.0000	
3.4540	1.7270	-0.3112	1.9756	-1.7270	-0.5000	-14.9080	-16.6677	7.4540	3.9756	7.1429	2.0000	
3.7680	1.8840	-0.6162	1.9027	-1.8840	-0.5000	-15.5361	-17.3698	7.7680	3.9027	7.1518	2.0000	
4.0820	2.0410	-0.9062	1.7829	-2.0410	-0.5000	-16.1641	-18.0720	8.0820	3.7829	7.1759	2.0000	
4.3960	2.1980	-1.1738	1.6193	-2.1980	-0.5000	-16.7921	-18.7741	8.3960	3.6193	7.2222	2.0000	
4.7100	2.3550	-1.4125	1.4159	-2.3550	-0.5000	-17.4201	-19.4762	8.7100	3.4159	7.2975	2.0000	
5.0240	2.5120	-1.6166	1.1776	-2.5120	-0.5000	-18.0481	-20.1784	9.0240	3.1776	7.4075	2.0000	
5.3380	2.6690	-1.7808	0.9104	-2.6690	-0.5000	-18.6761	-20.8805	9.3380	2.9104	7.5572	2.0000	
5.6520	2.8260	-1.9012	0.6207	-2.8260	-0.5000	-19.3041	-21.5826	9.6520	2.6207	7.7508	2.0000	
5.9660	2.9830	-1.9749	0.3158	-2.9830	-0.5000	-19.9321	-22.2847	9.9660	2.3158	7.9911	2.0000	
6.2800	3.1400	-2.0000	0.0031	-3.1400	-0.5000	-20.5601	-22.9869	10.2800	2.0031	8.2800	2.0000	
6.6003	3.3002	-1.9749	-0.3158	-3.3002	-0.5000	-21.2007	-23.7031	10.6003	1.6842	8.6254	2.0000	
6.9143	3.4572	-1.9012	-0.6207	-3.4572	-0.5000	-21.8287	-24.4052	10.9143	1.3793	9.0131	2.0000	
7.2283	3.6142	-1.7808	-0.9104	-3.6142	-0.5000	-22.4567	-25.1073	11.2283	1.0896	9.4475	2.0000	
7.5423	3.7712	-1.6166	-1.1776	-3.7712	-0.5000	-23.0847	-25.8094	11.5423	0.8224	9.9258	2.0000	
7.8563	3.9282	-1.4125	-1.4159	-3.9282	-0.5000	-23.7127	-26.5116	11.8563	0.5841	10.4438	2.0000	
8.1703	4.0852	-1.1738	-1.6193	-4.0852	-0.5000	-24.3407	-27.2137	12.1703	0.3807	10.9965	2.0000	
8.4843	4.2422	-0.9062	-1.7829	-4.2422	-0.5000	-24.9687	-27.9158	12.4843	0.2171	11.5782	2.0000	
8.7983	4.3992	-0.6162	-1.9027	-4.3992	-0.5000	-25.5967	-28.6180	12.7983	0.0973	12.1821	2.0000	

Figure 4.3. The Graphs tab (top) and the Tables tab (bottom) of the benchmark.

5 Conventions of the model

5.1 Units and signs

- Lengths in metres, forces in kilonewtons, angles in degrees in the input and in radians in the tables where the column says so. Unit weights in kN/m³, surcharges in kN/m².
- Load densities in kN per metre of *projected* length s, unless the input states another measure (Section 5.4).
- The axes $\hat{\mathrm{e}}_1$, $\hat{\mathrm{e}}_2$ are horizontal and $\hat{\mathrm{e}}_3$ points upward. A downward load has $p_3 < 0$.
- Tension is positive. A compressed line has $S \le 0$ and $N \le 0$. In the plane tasks the thrust $H = -F_1$ is positive in compression.
- Reactions are forces applied *to the line* by its supports.

5.2 Geometry of a spatial path

A spatial path is described by its projected arc length s, the length of its plan:

$$\mathrm{x}^*(s) = \mathrm{x}(s) + f(s)\,\hat{\mathrm{e}}_3, \qquad \hat{\mathrm{t}} = \mathrm{x}_{,s}, \qquad \hat{\mathrm{n}} = \hat{\mathrm{e}}_3 \times \hat{\mathrm{t}}, \qquad \hat{\mathrm{t}}_{,s} = \rho\,\hat{\mathrm{n}}, \tag{5.1}$$

where $\mathrm{x}(s)$ is the plan, $f(s)$ the elevation, $\hat{\mathrm{t}}$ the projected unit tangent, $\hat{\mathrm{n}}$ its left normal and ρ the projected curvature. The grade is $f_{,s}$, and $J = \sqrt{1 + f_{,s}^2}$ converts projected length into actual length. The path runs in the direction of increasing s: its start is $s = 0$ and its end $s = L$.

5.3 Forces

The force on the positive cut of the line is

$$\mathrm{F} = S\,\mathrm{u} = N\,\hat{\mathrm{T}}, \qquad \mathrm{u} = \hat{\mathrm{t}} + f_{,s}\,\hat{\mathrm{e}}_3, \qquad S = \mathrm{F}\cdot\hat{\mathrm{t}}, \qquad N = SJ, \tag{5.2}$$

where S is the thrust (the horizontal projection of the force along the plan tangent), N the axial force and $\hat{\mathrm{T}} = \mathrm{u}/J$ the unit tangent of the spatial path. The vertical component $V = F_3$ appears in the tables of the supported path. Every local result satisfies the balance $\mathrm{F}_{,s} + \mathrm{p} + \mathrm{r} = 0$, with p the load and r the reaction per projected metre.

5.4 Load measures

The vertical load is $p_3 = -w$, where w can be given:

- per projected metre, as a polynomial or a table in s, the measure of the article,
- per metre of actual length, as a polynomial, which suits a self-weight. Then $p_3 = -wJ$ per projected metre.

The horizontal load can be absent, a profile $h(s)$ in a fixed direction β measured from $\hat{\mathrm{e}}_1$ toward $\hat{\mathrm{e}}_2$ (a polynomial or a table), pseudo-static, $h = \gamma\,|p_3|$ along β, or given by two polynomial components $p_1(s)$, $p_2(s)$. The multiplier λ scales the horizontal load, so that $\lambda = 0$ removes it without changing its data. In the plane tasks the loads are per metre of the horizontal abscissa x, except a weight per metre of actual length (Chapter 10).

5.5 Point actions

A point action is a concentrated force $\mathrm{G} = (G_1, G_2, G_3)$ at the abscissa s. The program adds a node to the samples there, and every quantity has a value before and after the node. In the tables of stations the node takes two rows. On a supported path a point action takes a horizontal point reaction, and on a free curve the force jumps by $-\mathrm{G}$.

5.6 Resolution

The number of Samples along s (2001 by default, between 101 and 20 001) fixes the resolution of the integrals along the line, computed by the Simpson rule. The Output section (Fig. 5.1) sets the abscissa of the graphs, the angle θ when the path has one or the projected length s, and the number of Stations in the tables (41 by default). The stations do not affect the solution, only the rows of the tables and of the exports.

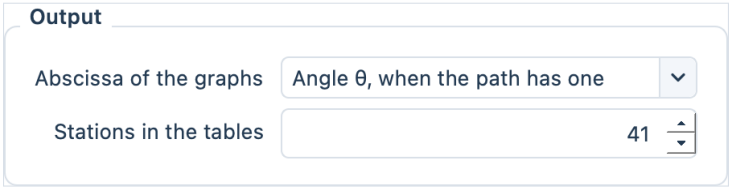


Figure 5.1. The Output section.

6 Supported path (assessment)

6.1 Purpose

The geometry of the path is prescribed, for instance the line of thrust of a helical stair or of a ramp, and the path leans on a lateral support, such as the wall of the stair well, that supplies horizontal reactions only ($r_3 = 0$). A preload $T = -S(0)$, the compressive end thrust at the start, completes the data. The program finds the thrust, the axial force and the reactions that the support must supply, and checks whether a wall with friction can supply them.

6.2 Input

Geometry of the path

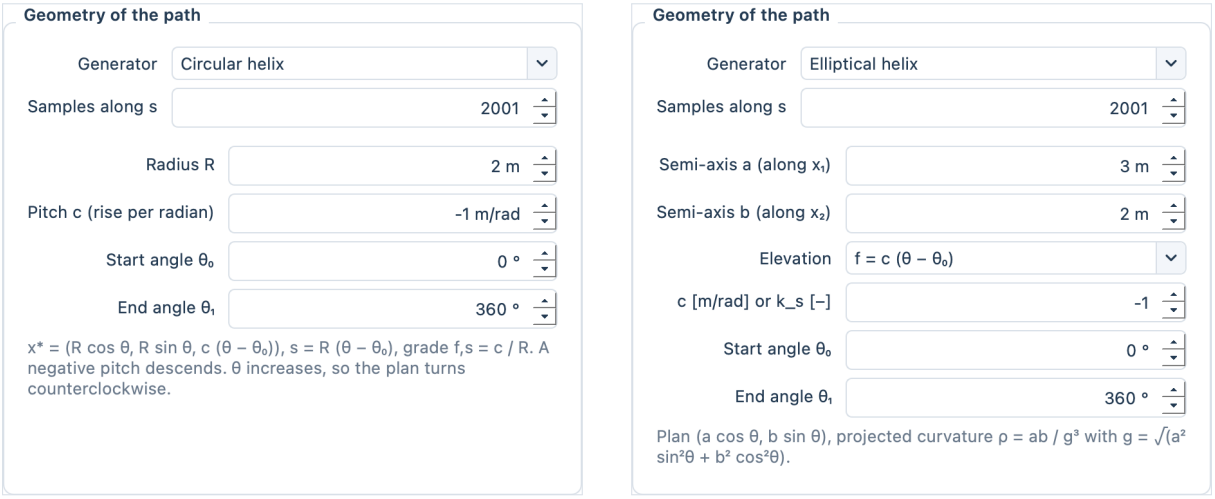


Figure 6.1. Geometry of the path: circular helix (left) and elliptical helix (right).

Field	Unit	Meaning
Generator		Circular helix, elliptical helix or smooth path through points.
Samples along s		Resolution of the integrals along s.
Radius R	m	Circular helix $(R\cos\theta, R\sin\theta, c(\theta-\theta_0))$, with $s = R(\theta-\theta_0)$ and grade c/R.
Pitch c (rise per radian)	m/rad	Negative for a path that descends as θ grows.
Start angle θ_0, end angle θ_1	°	The angular extent of the path, with $\theta_1 > \theta_0$.
Semi-axis a, b	m	Elliptical plan $(a\cos\theta, b\sin\theta)$.
Elevation		$f = c(\theta - \theta_0)$, or $f = k_s s$ with a constant grade.
c [m/rad] or k_s [–]		The rise per radian or the constant grade.
Points x_1, x_2, x_3	m	A cubic spline through the points, re-parametrized by the projected length. The plan must not stop or turn back.

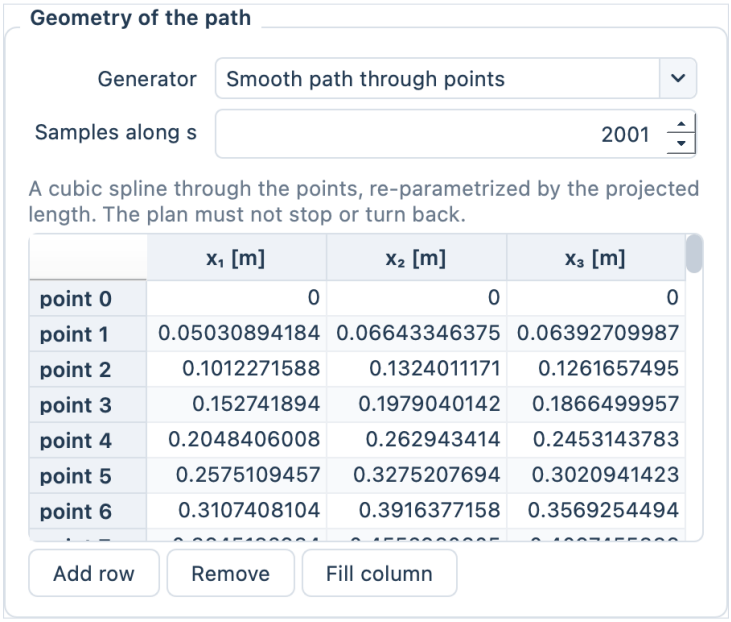


Figure 6.2. Geometry of the path: smooth path through points.

The supported path needs a non-zero grade everywhere, because the thrust follows from the vertical balance divided by the grade.

Distributed loads

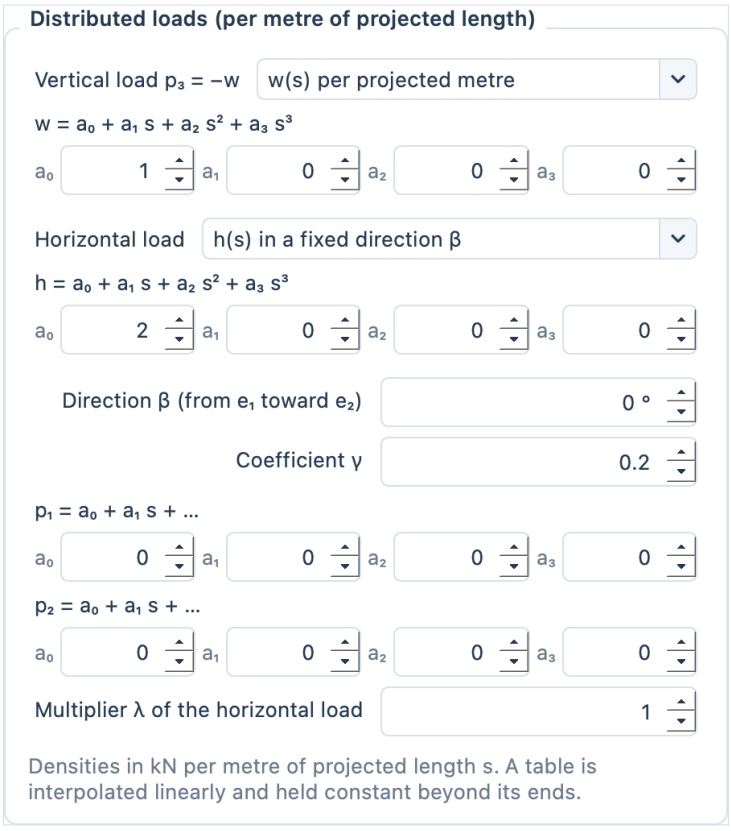


Figure 6.3. Distributed loads (per metre of projected length).

Field	Unit	Meaning
Vertical load $p_3 = -w$		$w(s)$ per projected metre, $w(s)$ per metre of actual length, or a table of $w(s)$ per projected metre.
$w = a_0 + a_1\, s + a_2\, s^2 + a_3\, s^3$	kN/m	Coefficients of w (or the table s, w).
Horizontal load		None, $h(s)$ in a fixed direction β, a table of $h(s)$, pseudo-static $h = \gamma\lvert p_3\rvert$ along β, or the components $p_1(s)$, $p_2(s)$.
$h = a_0 + \ldots$	kN/m	Coefficients of h.
Direction β	°	Direction of the horizontal load, from $\hat{e}_1$ toward $\hat{e}_2$.
Coefficient γ	–	Ratio of the pseudo-static horizontal load to the vertical load.
$p_1, p_2 = a_0 + \ldots$	kN/m	Components of the horizontal load.
Multiplier λ	–	Factor on the horizontal load.

Point actions

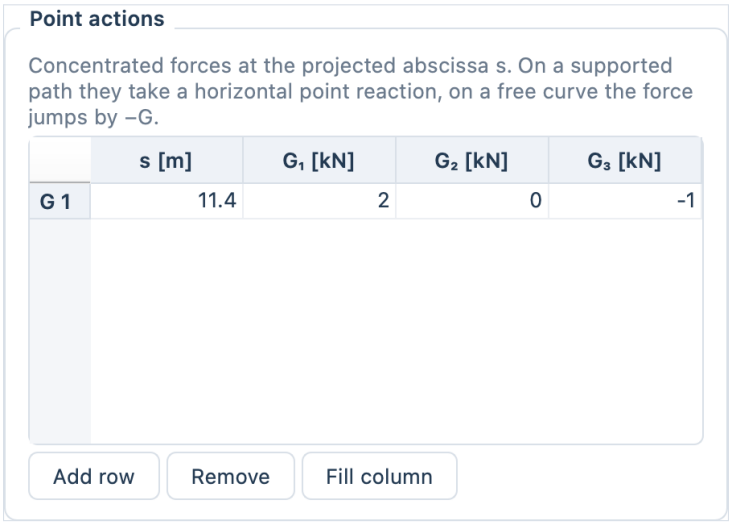


Figure 6.4. Point actions with one concentrated force.

Each row gives the abscissa s and the components G_1, G_2, G_3 in kN of a concentrated force.

Support of the line and contact with the wall

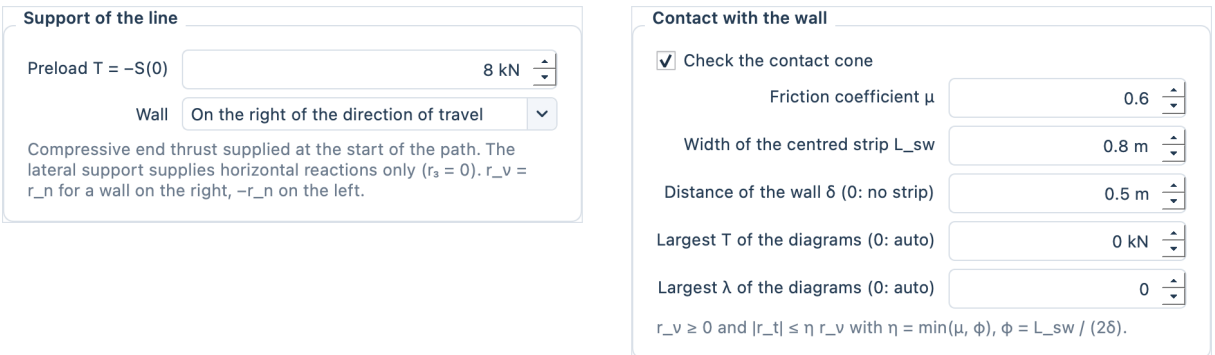


Figure 6.5. Support of the line (left) and Contact with the wall (right).

Field	Unit	Meaning
Preload T = −S(0)	kN	Compressive end thrust supplied at the start of the path.
Wall		On the right or on the left of the direction of travel.
Check the contact cone		Switches on the contact check and the Contact tab.
Friction coefficient μ	–	Friction between the line and the wall.
Width of the centred strip L_{sw}	m	Width of the strip of contact.
Distance of the wall δ	m	Distance of the strip from the wall. 0 removes the strip limit.
Largest T, largest λ of the diagrams	kN, –	Window of the diagram of admissible pairs (T, λ). 0 chooses it automatically.

6.3 Method

Vertical balance gives the vertical component of the force, $V = C_3 - P_3(s)$, with C = F(0) and P_3 the primitive of the vertical loads. Hence

$$S = \frac{V}{f_{,s}}, \qquad N = SJ, \qquad r_t = -S_{,s} - \mathrm{p}_h \cdot \hat{\mathrm{t}}, \qquad r_n = -\rho\, S - \mathrm{p}_h \cdot \hat{\mathrm{n}}, \tag{6.1}$$

where r_t and r_n are the tangential and normal components of the horizontal reaction and p_h the horizontal load. A horizontal load leaves the thrust unchanged and shifts the reaction by $-\mathrm{p}_h$: the dashed arrows of the plan show the reaction of the vertical loads alone.

A point action G at s_j, with no vertical support impulse, makes the thrust jump and requires a horizontal point reaction:

$$[S] = -\frac{G_3}{f_{,s}}, \qquad \mathrm{R}_h = \frac{G_3}{f_{,s}}\,\hat{\mathrm{t}} - \mathrm{G}_h. \tag{6.2}$$

6.4 Contact with the wall

The wall presses the line with the normal reaction $r_\nu = r_n$ for a wall on the right of the direction of travel and $r_\nu = -r_n$ for a wall on the left. The reaction is admissible when it lies in the contact cone

$$r_\nu \geq 0, \qquad |r_t| \leq \eta\, r_\nu, \qquad \eta = \min(\mu, \varphi), \qquad \varphi = \frac{L_{sw}}{2\delta}, \tag{6.3}$$

where φ limits the tangential reaction that a centred strip of width L_{sw} at the distance δ of the wall can carry. With $\delta = 0$ the strip limit is removed. The contact margin $\min(r_\nu,\ \eta r_\nu - |r_t|)$ is non-negative exactly inside the cone.

Every reaction is affine in the preload T and in the load multiplier λ. The program therefore gives in closed form:

- the interval of admissible preloads at the given λ, and the side of the cone and the station that govern it,
- the interval of admissible multipliers at the given T, when there is a horizontal load,
- the convex region of admissible pairs (T, λ),
- the admissible horizontal loads turned to every direction β at the given preload,
- the minimum preload against the size of the horizontal load, along its direction and along the directions turned by 90° and 180°,
- for a descending circular helix with constant loads per projected metre, a wall on the right and no point actions, the exact minimum preload of Appendix A of the article.

> **Caution** A violated contact cone means that the assumed wall cannot supply the required reaction. It does not mean that the line loses compression, and the bounds on T and λ are not collapse loads.

6.5 Results

Summary. Geometry, projected and actual length, drop, preload, ranges of S, N, r_t and r_n, end forces, compression, wall, global balance and, with the contact check, the cone, the least margin, the admissible preloads and multipliers and the exact bound when it applies.

Views. Plan with loads and reactions and Developed elevation.

Graphs. Thrust and axial force and Required reactions.

Contact. Contact margin, Contact cone, Admissible preload and load multiplier, Admissible horizontal loads at T = … and Minimum preload along the load ray.

Tables. Stations (s, θ, position, grade, S, N, V, r_t, r_n, loads and margin), Point actions (jump of S, point reaction and its cone margin), Balance and Summary.

6.6 Worked example: the circular benchmark

Load **Examples › Supported: Figs. 11 and 12, circular benchmark with contact**. The path is a complete descending turn of a circular helix of radius 2 m and pitch -1 m/rad, under a vertical load of 1 kN per projected metre and a horizontal load of 2 kN/m along $\hat{e}_1$, with a preload of 8 kN and a wall on the right. The contact cone has $\mu = 0.6$ and a strip of 0.8 m at 0.5 m from the wall. The summary reads:

Quantity	Value
Geometry	Circular helix R = 2 m, c = -1 m/rad, θ from 0° to 360°
Projected length L	12.5664 m
Length of the spatial path	14.0496 m
Drop $x_3(0) - x_3(L)$	6.28319 m
Preload T = −S(0)	8 kN
Thrust S	-8 kN to -33.1327 kN
Axial force N = S J	-8.94427 kN to -37.0435 kN
Force at the start F(0)	(0, -8, 4) kN
Force at the end F(L)	(-8.11518e-15, -33.1327, 16.5664) kN
Compression	$S \leq 0$ along the whole line
Tangential reaction r_t	6.16234e-07 kN/m to 4 kN/m
Normal reaction r_n	6 kN/m to 18.5664 kN/m
Wall	on the right of the direction of travel ($r_\nu = r_n$)
Global balance (normalised)	force 9.0e-13, moment 5.5e-13
Contact cone	$\eta = \min(\mu, \varphi) = 0.6$ with $\mu = 0.6$, $\varphi = 0.8$
Contact margin	min 0.284956 kN/m at s = 3.14 m ($\theta = 1.57$ rad): satisfied
Admissible preloads at λ = 1	7.05015 kN ≤ T ≤ unbounded, governed by cone side $r_t \leq \eta\, r_\nu$ at s = 3.14 m, $\theta = 1.57$ rad
Admissible multipliers at T = 8 kN	-0.333333 ≤ λ ≤ 1.13945, governed by cone side $r_t \leq \eta\, r_\nu$ at s = 3.284 m, $\theta = 1.642$ rad
Exact minimum preload (Appendix A)	7.05015 kN, active at $\theta - \theta_0 = 1.5708$ rad

The contact margin is positive, so the wall can supply the reactions. The admissible preloads start at the exact bound of Appendix A, and the horizontal load could grow by the factor printed as the upper

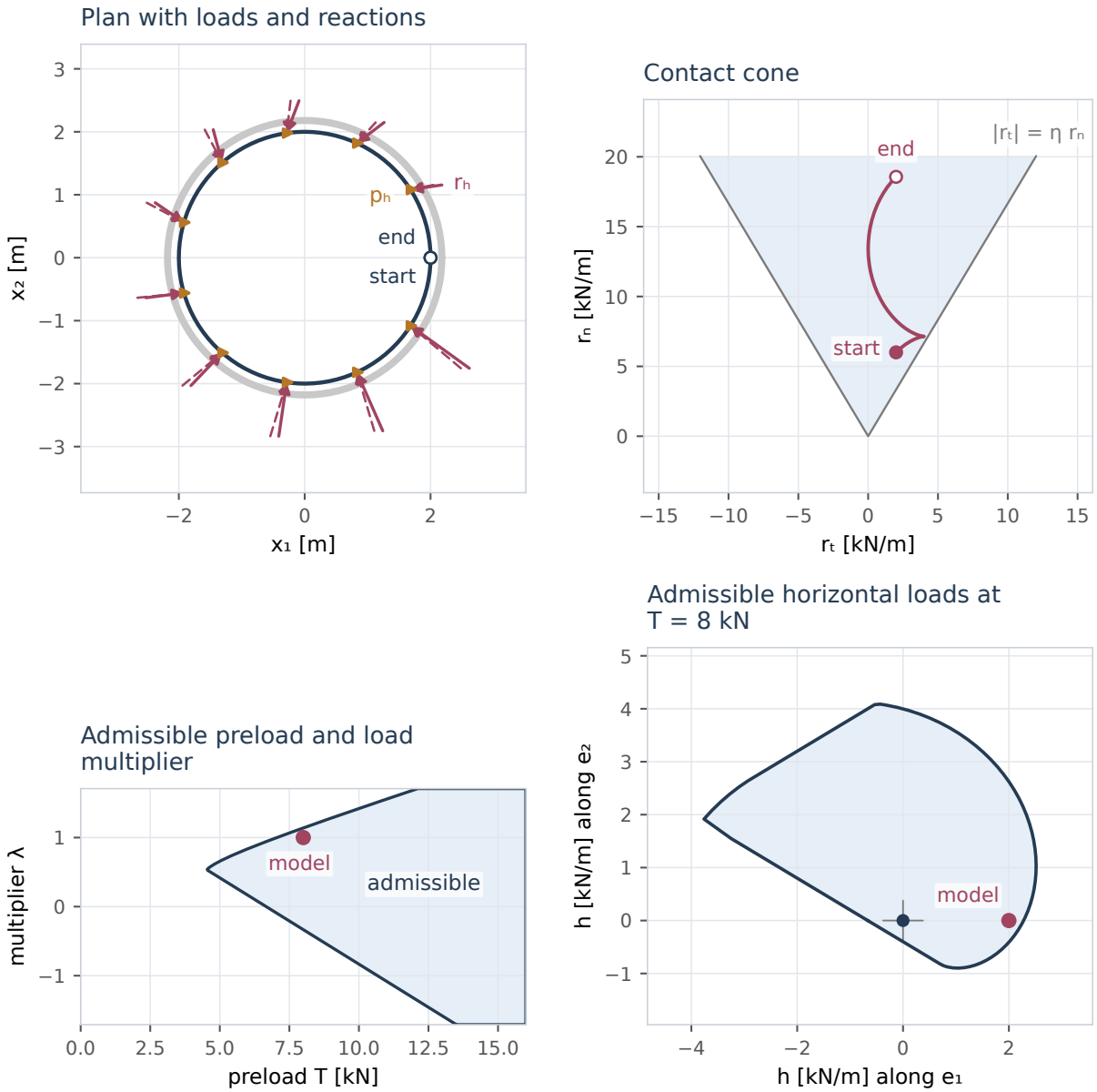


Figure 6.6. Drawings of the benchmark exported by Funis: plan with loads and reactions, contact cone, admissible pairs of preload and load multiplier, admissible horizontal loads at the given preload.

admissible multiplier before the cone is violated. The example **Supported: Fig. 11, contact violated at h/w = 4** doubles the horizontal load: the warning

- The required reaction leaves the contact cone at s = 3.699 m (θ = 1.849 rad) (margin -1.887 kN/m): the assumed wall cannot supply it, although the line may remain in compression.

appears above the summary, and Fig. 6.7 shows it.

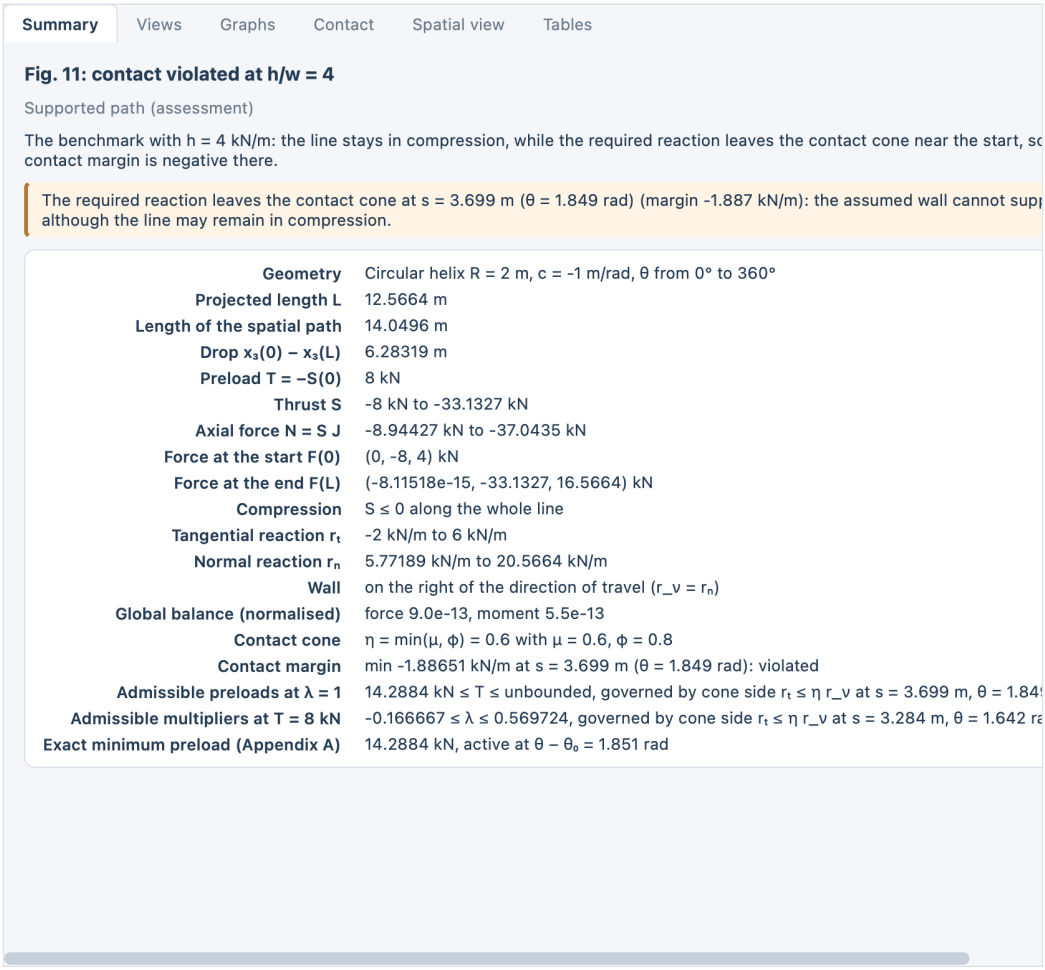


Figure 6.7. A warning in the Summary tab: the reaction leaves the contact cone.

7 Path without reactions

7.1 Purpose

Without reactions the force along the line is $F = C - P(s)$, with C the force at the start and P the primitive of the distributed loads and of the point forces. A prescribed path is an axial equilibrium only if F is parallel to u everywhere. This task checks that condition for given loads, or finds the loads that satisfy it.

7.2 Input

The sections Geometry of the path and Output are those of Chapter 6. The section Path without reactions (Fig. 7.1) selects the task, and Distributed loads and Point actions appear only to check given loads.

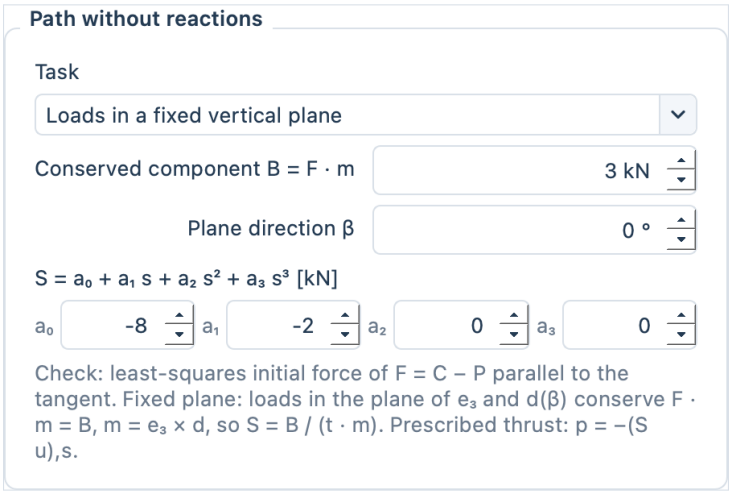


Figure 7.1. The section Path without reactions.

Field	Unit	Meaning
Task		Check the loads below, loads in a fixed vertical plane, or loads for a prescribed thrust $S(s)$.
Conserved component B = F · m	kN	For loads in a fixed vertical plane.
Plane direction β	°	Direction $\hat{\mathrm{d}}(\beta)$ of the vertical plane of the loads.
$S = a_0 + a_1 s + \ldots$	kN	The prescribed thrust.

7.3 Method

Check the loads. The program finds the initial force C that makes $F = C - P$ closest to the tangent in the least-squares sense, and reports the largest transverse force $|F - Su|$. A transverse force that is not small, compared with the force itself, disproves the equilibrium for every C: a lateral support or a different geometry is needed.

Loads in a fixed vertical plane. Loads in the plane spanned by $\hat{\mathrm{e}}_3$ and $\hat{\mathrm{d}}(\beta)$ conserve the component $F \cdot \hat{\mathrm{m}} = B$ along $\hat{\mathrm{m}} = \hat{\mathrm{e}}_3 \times \hat{\mathrm{d}}$, so that $S = B/(\hat{\mathrm{t}} \cdot \hat{\mathrm{m}})$ and the compatible loads follow from $p = -F_{,s}$. Compression needs $\hat{\mathrm{t}} \cdot \hat{\mathrm{m}}$ of opposite sign to B, which restricts the path to part of a half-turn.

Loads for a prescribed thrust. The loads are $p = -(Su)_{,s}$, in general with three components.

7.4 Results

The summary gives the verdict or the compatible loads, the ranges of S and N, the compression and the global balance. The views are the plan with the horizontal loads and the developed elevation, and the graphs show the forces, the loads and, for a check, Transverse force of the best fit. A part of the path in tension is reported with a warning.

7.5 Worked example

Examples › Without reactions: Fig. 3, path that is not an axial equilibrium perturbs the elevation of a funicular curve. The check finds:

Quantity	Value
Geometry	Smooth path through 61 points
Projected length L	5 m
Best initial force C	(-0.572289, -1.91275, -2.00582) kN
Largest transverse force |F − S u|	0.815184 kN at s = 0 m
Relative to the force	1.34e-01
Verdict	no initial force makes the path an axial equilibrium of these loads
Thrust S	-2.12946 kN to -4.25385 kN
Axial force N	-2.4791 kN to -5.36895 kN
Compression	S ≤ 0 along the whole path
Global balance (normalised)	force 0.0e+00, moment 1.2e-02

with the warning

- The assigned path is not compatible with these loads without reactions: the force F = C − P has a transverse part of up to 0.8152 kN for every C. A lateral support (supported path) or a different geometry is needed.

The example that follows it in the menu solves the same check on the funicular itself, which passes.

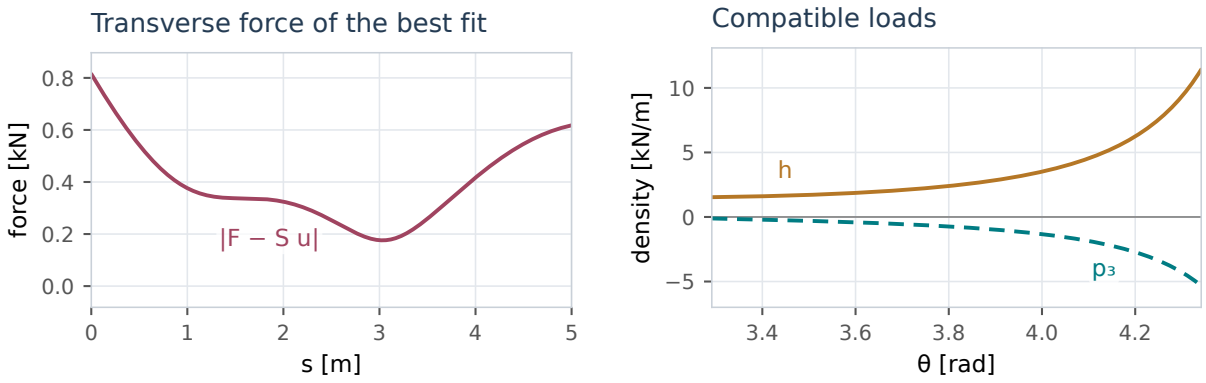


Figure 7.2. Transverse force of the best fit for the perturbed path (left), and the compatible loads of the exceptional helical arc of Fig. 8 of the article (right).

8 Unknown geometry (form-finding)

8.1 Purpose

The loads are given as functions of the projected length and the curve is unknown. Starting from a point and from the initial force, the program integrates the curve that carries the loads in pure compression. It can also find the initial force that takes the curve to a prescribed second support.

8.2 Input

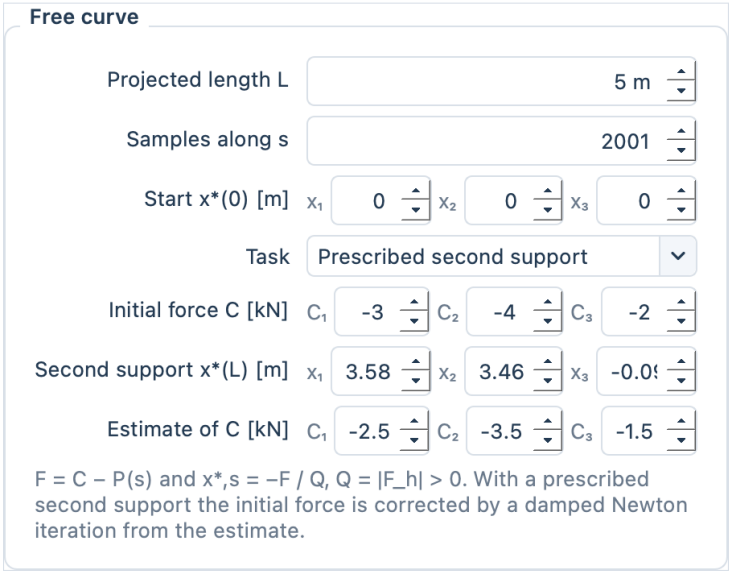


Figure 8.1. The section Free curve with a prescribed second support.

Field	Unit	Meaning
Projected length L	m	Length of the plan of the curve.
Samples along s		Resolution of the integration.
Start x*(0)	m	The first support.
Task		Prescribed initial force, or prescribed second support.
Initial force C	kN	C = F(0), for a prescribed initial force.
Second support x*(L)	m	The target, for a prescribed second support.
Estimate of C	kN	Starting value of the correction, for a prescribed second support.

The sections Distributed loads and Point actions are those of Chapter 6.

8.3 Method

The force is $\mathrm{F} = \mathrm{C} - \mathrm{P}(s)$ and the curve follows by quadrature from

$$\mathrm{x}^*_{,s} = -\frac{\mathrm{F}}{Q}, \qquad Q = |\mathrm{F}_h| > 0, \qquad S = -Q, \qquad N = -|\mathrm{F}|, \tag{8.1}$$

where F_h is the horizontal part of the force. With a weight per metre of actual length the force and the position are integrated together. With a prescribed second support, the initial force is corrected by a damped Newton iteration with the exact endpoint Jacobian (finite differences for a weight per actual length), starting from the estimate. Every iterate keeps $Q > 0$.

The graphs give the curvature and the torsion of the curve,

$$\kappa = \frac{Q\,|\mathrm{F} \times \mathrm{p}|}{|\mathrm{F}|^3}, \qquad \tau = -\frac{Q\,\mathrm{F} \cdot (\mathrm{p} \times \mathrm{p}_{,s})}{|\mathrm{F} \times \mathrm{p}|^2}. \tag{8.2}$$

8.4 Results

The summary gives the end point, the end force, the length of the curve, the ranges of S and N, the least horizontal force Q, the distance from the plane of C and p(0) (zero for parallel loads) and, for constant loads, the difference from the closed form of Appendix B of the article. With a prescribed second support it adds the target, the number of force updates with the final mismatch, and the eigenvalues of the plan block K of the endpoint Jacobian. The views are the plan and the developed elevation, with Elevations of the successive estimates for a second support. The graphs show the forces, the curvature and torsion, the distance from the plane and the endpoint correction.

8.5 Worked example

Examples › Form-finding: Fig. 6, prescribed second support prescribes the second support of case C of the article and starts from a deliberately wrong estimate:

Quantity	Value
Projected length L	5 m
Start x*(0)	(0, 0, 0) m
Initial force C = F(0)	(-3, -4, -2) kN
End x*(L)	(3.58655, 3.46958, -0.0939623) m
Force at the end F(L)	(-9, -7.75, 3) kN
Length of the curve	5.09027 m
Thrust S = −Q	-5 kN to -11.877 kN
Axial force N = −|F|	-5.38517 kN to -12.25 kN
Least horizontal force Q	5 kN
Target x*(L)	(3.58655, 3.46958, -0.0939623) m
Force updates	4, final mismatch 5.03e-14 m
Eigenvalues of K	0.00304553 and 0.693717 m/kN
Distance from the plane of C and p(0)	up to 0.179997 m
Global balance (normalised)	force 3.7e-17, moment 3.2e-15

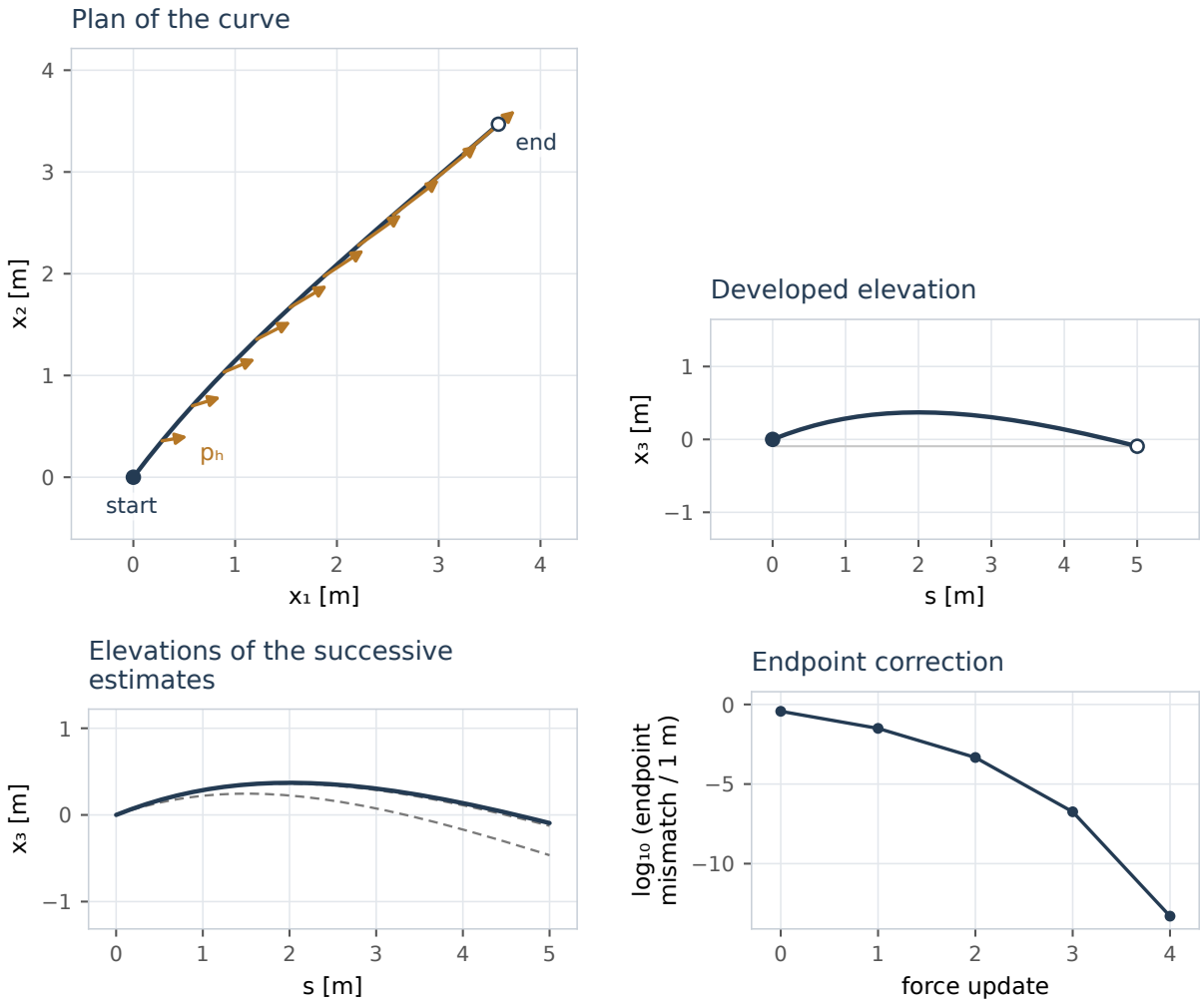


Figure 8.2. Form-finding: plan and developed elevation of case C (top), and the successive estimates and the endpoint mismatch of the correction for a prescribed second support (bottom).

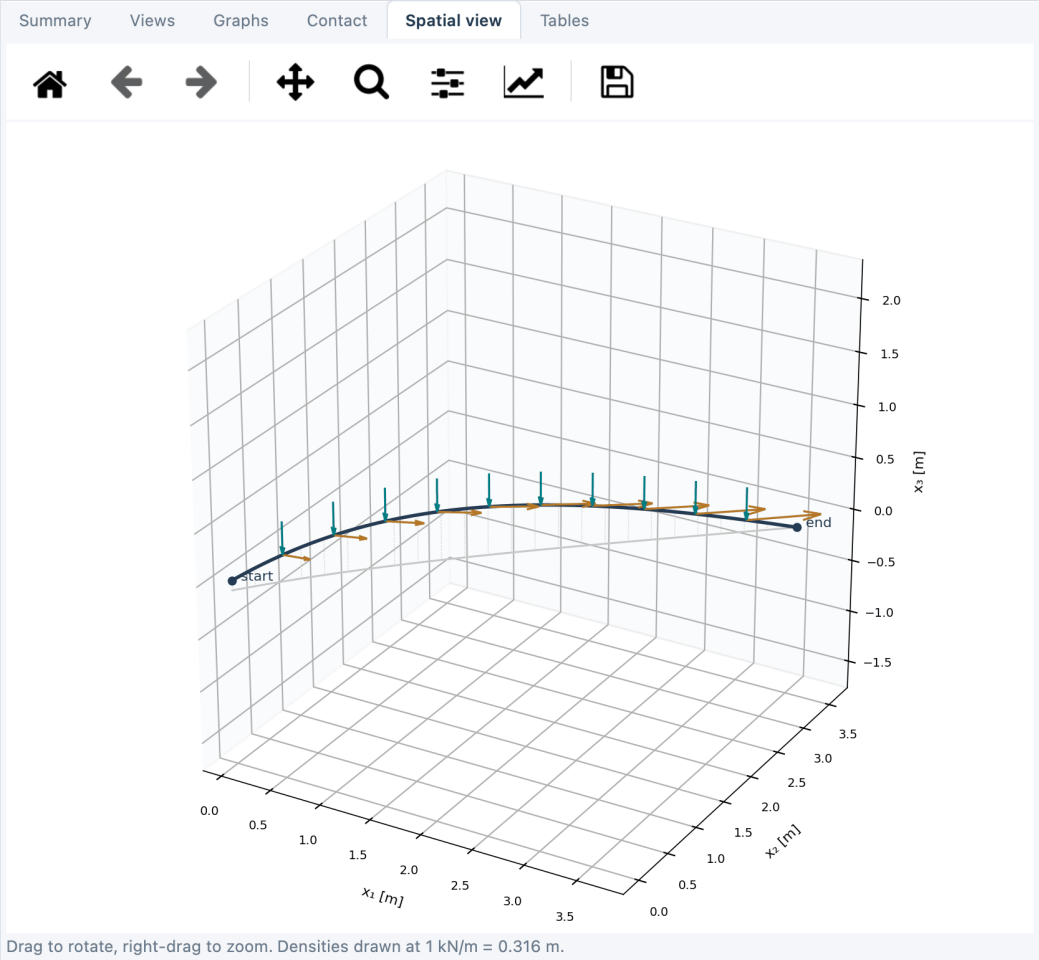


Figure 8.3. The Spatial view of case C. Drag to rotate, drag with the right button to zoom.

9 Node under a concentrated force

9.1 Purpose and input

At a node where a concentrated force G and possibly a point reaction R act, the force of the line jumps. The task gives the force after the node and the change of direction of the line.

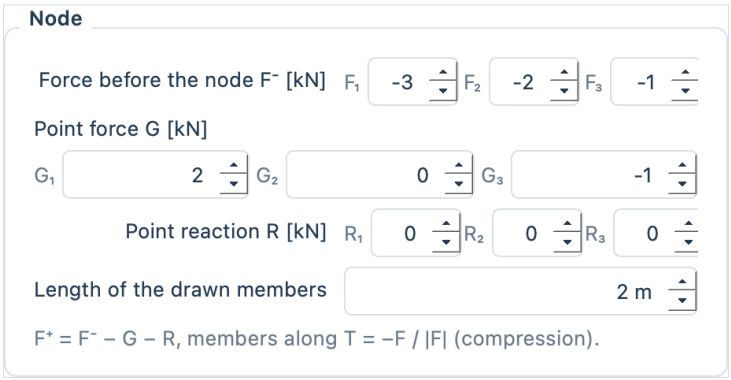


Figure 9.1. The section Node.

Field	Unit	Meaning
Force before the node F^-	kN	The force of the line before the node.
Point force G	kN	The concentrated force.
Point reaction R	kN	An optional point reaction.
Length of the drawn members	m	Length of the members in the drawings.

9.2 Method and results

$$F^+ = F^- - G - R, \qquad \hat{T}^\pm = -\frac{F^\pm}{|F^\pm|}. \tag{9.1}$$

The summary gives F^+, the axial forces N^- and N^+, the tangents, the change of tangent in degrees and the closure of the force polygon. The views are the node in plan, in elevation on x_1x_3 and the force polygon, drawn in the plane of the forces. For **Examples › Node: Fig. 7, concentrated multidirectional force**:

Quantity	Value
Force before the node F^-	(-3, -2, -1) kN
Point force G	(2, 0, -1) kN
Point reaction R	(0, 0, 0) kN
Force after the node $F^+ = F^- - G - R$	(-5, -2, 0) kN
Axial forces N^-, N^+	-3.74166 kN, -5.38516 kN
Tangent before T^-	(0.801784, 0.534522, 0.267261)
Tangent after T^+	(0.928477, 0.371391, -0)
Change of tangent	19.4463 °
Closure of the force polygon	0.0e+00 kN

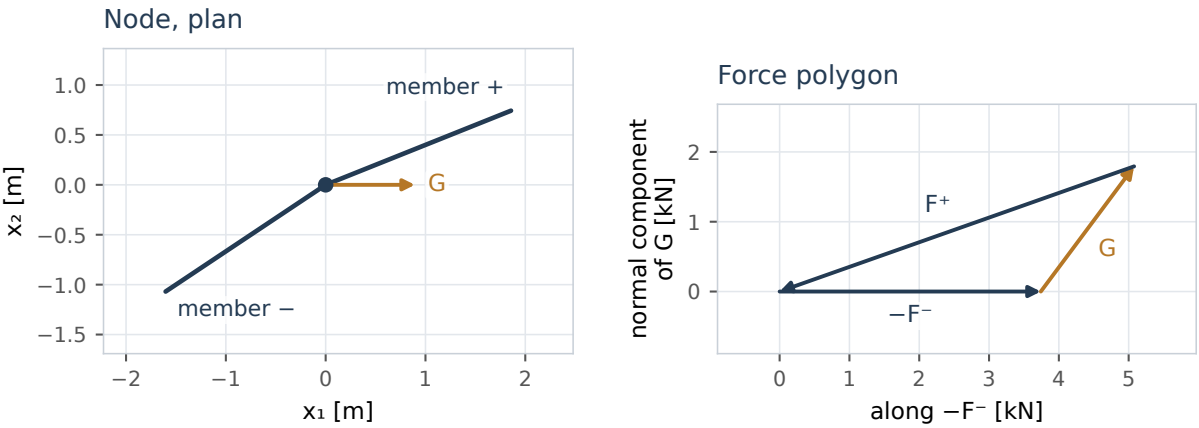


Figure 9.2. The node of Fig. 7 of the article in plan, and its force polygon.

10 Plane funicular

10.1 Purpose

The line lies in the vertical plane of $\hat{e}_1$ and $\hat{e}_3$, between the supports $A = (0, 0)$ and $B = (L, z_B)$, and is described by its height $z = f(x)$ over the horizontal abscissa x. The program finds the line that carries the loads in compression: the catenary under a weight per metre of actual length, the parabola under a load per metre of x, the funicular polygon of point loads, and their combinations with horizontal loads.

10.2 Input

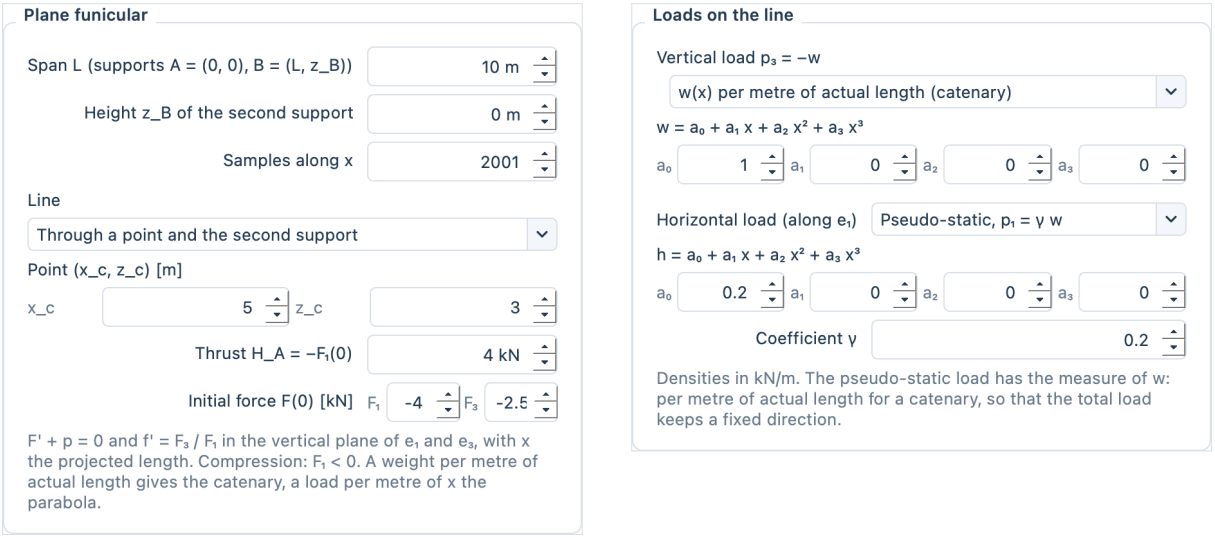


Figure 10.1. The sections Plane funicular (left) and Loads on the line (right).

Field	Unit	Meaning
Span L	m	Horizontal distance of the supports.
Height z_B of the second support	m	Height of B above A.
Samples along x		Resolution of the integration.
Line		Through a point and the second support, with a prescribed thrust and the second support, or from a prescribed initial force.
Point (x_c, z_c)	m	A point of the line, with $0 < x_c < L$.
Thrust $H_A = -F_1(0)$	kN	The thrust at A, positive in compression.
Initial force F(0)	kN	The force at A. The line then ends where the loads take it.
Vertical load $p_3 = -w$	kN/m	$w(x)$ per metre of actual length (catenary), per metre of x (parabola), or a table per metre of x.
Horizontal load (along e_1)	kN/m	None, $p_1 = h(x)$ per metre of x, or pseudo-static $p_1 = \gamma w$ in the measure of w.
Point loads x, G_1, G_3	m, kN	Concentrated forces on the line. $G_3 < 0$ acts downward.

10.3 Method

Without reactions the force is $\mathrm{F} = \mathrm{C} - \mathrm{P}(x)$, with P the primitive of the loads and of the point loads, and the line follows $f_{,x} = F_3/F_1$. Compression needs $F_1 < 0$ everywhere. The thrust is $H = -F_1$, the axial force $N = -|\mathrm{F}|$, and the reactions are $\mathrm{R}_A = -\mathrm{F}(0)$ and $\mathrm{R}_B = \mathrm{F}(L)$. A load per metre of actual length is multiplied by $J = |\mathrm{F}|/|F_1|$, and the force and the line are integrated together. A horizontal load makes the thrust vary along the line, and where F_1 would reach zero the line turns vertical and the program stops with a message. A point load makes the force jump by $-\mathrm{G}$ and the line has a corner there.

Where a closed form exists the summary gives its largest distance from the integrated line: the parabola (constant w per metre of x), the catenary (constant w per metre of actual length), the logarithmic branch of Appendix B of the article (constant w and h per metre of x) and the catenary with an inclined axis (constant w per actual length with the pseudo-static load).

10.4 Results and worked example

The summary gives the loads, the selected line, the initial force, the thrusts and reactions at A and B, the highest point, the length, the range of N, the height reached at $x = L$ against the target, the closed form when it applies and the global balance. For **Examples › Plane: catenary through a point**:

Quantity	Value
Span L	10 m
Second support B	(10, 0) m
Vertical load	w = 1 kN/m per metre of actual length (catenary)
Horizontal load	none
Line	through (5, 3) m and the second support
Initial force F(0)	(-4.59447, -6.04705) kN
Thrust at A, H_A = $-F_1(0)$	4.59447 kN
Thrust at B, H_B = $-F_1(L)$	4.59447 kN
Reaction at A, R_A = −F(0)	(4.59447, 6.04705) kN
Reaction at B, R_B = F(L)	(-4.59447, 6.04705) kN
Highest point	(5, 3) m
Length of the line	12.0941 m
Axial force N = −|F|	-4.59447 kN to -7.59447 kN
Height reached at x = L	-4.81837e-14 m (target 0 m)
Closed form against the integration	4.70e-11 m (catenary, Appendix B)
Global balance (normalised)	force 8.7e-15, moment 1.9e-13

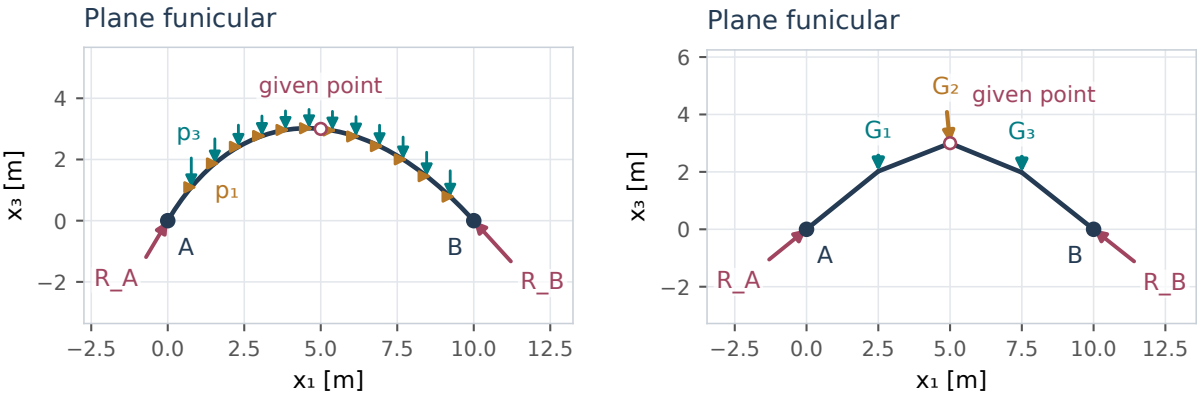


Figure 10.2. A catenary under its weight and a pseudo-static horizontal action (left) and a funicular polygon of point loads (right), with the reactions.

11 Arch assessment (thrust line)

11.1 Purpose

The task assesses a masonry arch of finite thickness by its thrust lines, under the classical hypotheses of Heyman [3]: the masonry has no tensile strength, an unlimited compressive strength, and sliding between voussoirs does not occur unless a friction coefficient is given. By the safe theorem, if one thrust line lies within the arch under the given loads, the arch does not collapse under those loads.

11.2 Input

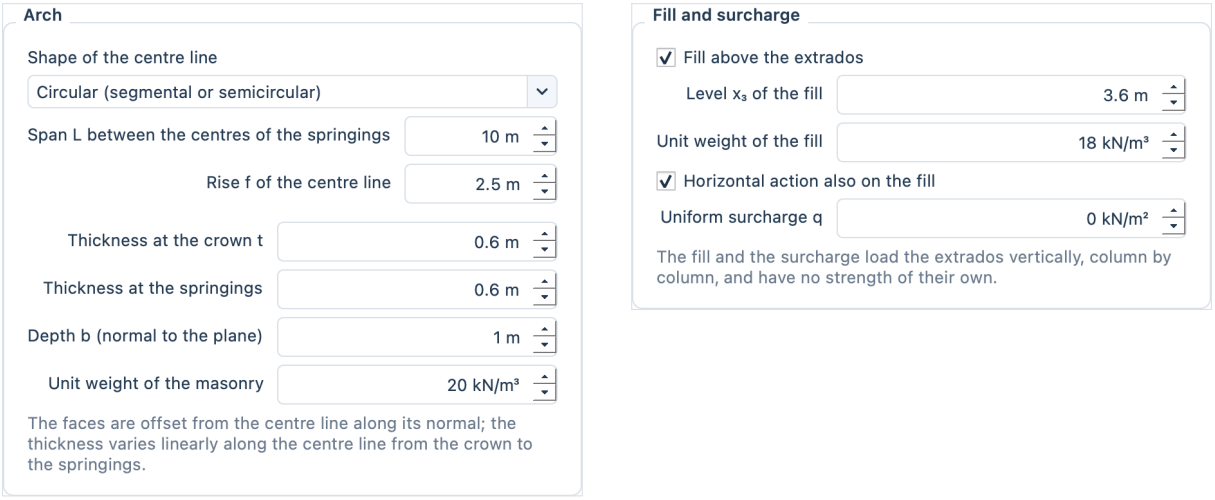


Figure 11.1. The sections Arch (left) and Fill and surcharge (right).

Field	Unit	Meaning
Shape of the centre line		Circular (segmental or semicircular), pointed (two-centred), elliptical, parabolic, catenary, or through points (x, z).
Span L	m	Distance of the centres of the springings, at $(0, 0)$ and $(L, 0)$.
Rise f	m	Rise of the centre line.
Thickness at the crown t, at the springings	m	The thickness varies linearly along the centre line between them. The faces are offset from the centre line along its normal.
Depth b	m	Depth normal to the plane, which multiplies every area.
Unit weight of the masonry	kN/m^3	
Fill above the extrados		Switches on the fill.
Level x_3 of the fill, unit weight	m, kN/m^3	The fill loads the extrados vertically, column by column, and has no strength of its own.
Horizontal action also on the fill		Applies the horizontal action to the weight of the fill too.
Uniform surcharge q	kN/m^2	Vertical load on the extrados.

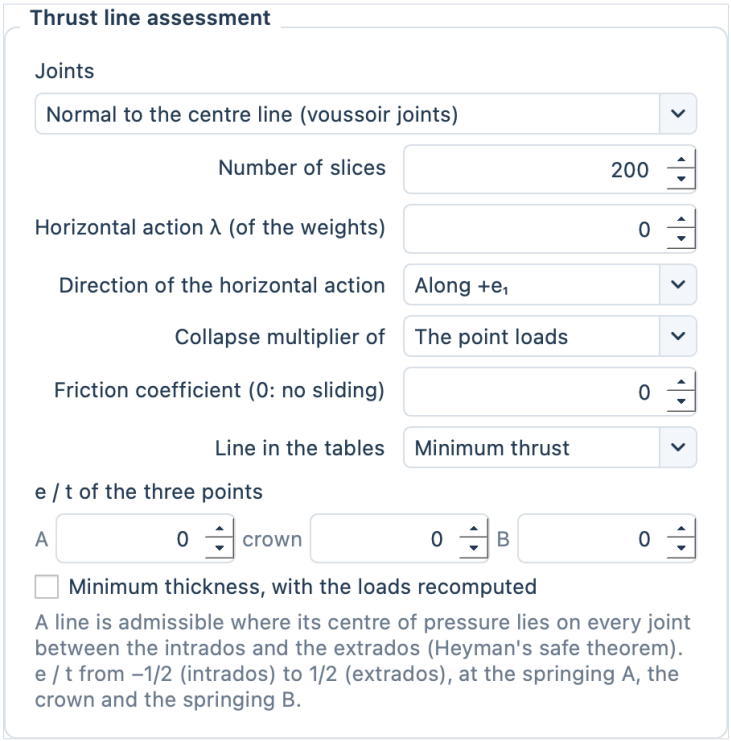


Figure 11.2. The section Thrust line assessment.

Field	Unit	Meaning
Joints		Normal to the centre line (voussoir joints) or vertical sections.
Number of slices		The arch is cut into slices by the joints. Each slice carries its weight at its centroid.
Horizontal action λ	–	A fraction of the weights applied horizontally at the same centroids.
Direction of the horizontal action		Along $+\hat{e}_1$ or $-\hat{e}_1$.
Collapse multiplier of		The horizontal action, the point loads, or none.
Friction coefficient	–	With $\mu > 0$ every joint also needs $\lvert V\rvert \leq \mu \lvert N\rvert$. With 0 sliding is excluded.
Line in the tables		Minimum thrust, maximum thrust, or the line through three points.
e / t of the three points	–	Eccentricities at the springing A, at the crown and at the springing B, from $-1/2$ (intrados) to $1/2$ (extrados).
Minimum thickness		Computes the least thickness, with the weights recomputed.

The section Point loads gives concentrated forces (x, G_1, G_3) on the extrados at the abscissa x.

11.3 Method

A thrust line is the line of the centres of pressure of the resultant forces on the joints. The force on joint j is $\mathrm{F}_j = \mathrm{C} - \mathrm{P}_j$, with C the force at the springing A and P_j the sum of the loads of the slices before the joint, and the moment follows in the same way, so that every line is fixed by three numbers. The line is admissible where its centre of pressure lies on every joint between the intrados and the extrados and the force compresses the joint. These conditions are linear in the three numbers, and the program solves them as linear programs:

Minimum and maximum thrust. The admissible lines with the least and the largest horizontal thrust H_A, with the joints where they touch a face.

Line through three points. The line with the given eccentricities at A, at the crown and at B. It is reported even when it leaves the arch, with a warning.

Geometric safety factor. The joints are shrunk about their centres until a line just fits, with the same loads. The factor is the actual thickness over the thickness of this reduced arch [4]. A factor below 1 means that no line lies within the arch.

Collapse multiplier. The largest multiplier of the horizontal action or of the point loads for which an admissible line exists. The line touches the faces at the hinges of the mechanism. With four hinges and no friction limit, the summary checks the multiplier by the virtual work of the mechanism through them.

Minimum thickness. The least factor on every thickness for which a line fits, with the weights recomputed for the thinner arch. For the semicircular arch under its own weight on normal joints this is the classical $t/R = 0.1075$, with hinges near 54.5° from the crown.

Friction. With a coefficient $\mu > 0$ the conditions $|V| \leq \mu|N|$ join the linear programs. With 0 the summary gives the friction that the selected line needs.

11.4 Results

Views. Thrust lines in the arch, Loads and reactions of the selected line, Collapse with the hinges of the mechanism, Geometric safety factor with the reduced arch, and Arch of minimum thickness.

Graphs. Eccentricity of the centre of pressure and Normal force at the joints.

Admissible thrusts. The range of H_A of the lines in the arch against the multiplier of the variable loads. It closes at the collapse multiplier.

Tables. Joints (position, angle, length, forces, e/t, moment and centre of pressure of the selected line), Eccentricities, Hinges, Admissible thrusts, Loads, Balance and Summary.

11.5 Worked examples

Examples › Arch: semicircular arch under a horizontal action computes the collapse multiplier of a horizontal action proportional to the weight of a semicircular arch with $t/R = 0.15$:

Quantity	Value
Arch	Circular arch, span 10 m, rise 5 m, radius of the centre line 5 m, thickness 0.75 m
Joints	201 joints normal to the centre line, 200 slices
Weight of the arch	235.619 kN (γ = 20 kN/m^3, depth 1 m)
Horizontal action	none
Sliding	excluded (Heyman: no sliding)
Minimum thrust H_A	41.4388 kN, H_B = 41.4388 kN; touches intrados at x = 1.095 m, 57.6° from the vertical; extrados at x = 5 m, 0° from the vertical; intrados at x = 8.905 m, 57.6° from the vertical
Maximum thrust H_A	53.4803 kN, H_B = 53.4803 kN; touches extrados at x = -0.375 m, 90° from the vertical; intrados at x = 1.781 m, 44.1° from the vertical; intrados at x = 8.219 m, 44.1° from the vertical; extrados at x = 10.38 m, 90° from the vertical
Geometric safety factor	1.3952: the joints shrunk about their centres to 0.7168 of their length still contain a line, with extrados at x = -0.375 m, 90° from the vertical; intrados at x = 1.216 m, 54.9° from the vertical; extrados at x = 5 m, 0° from the vertical; intrados at x = 8.784 m, 54.9° from the vertical; extrados at x = 10.38 m, 90° from the vertical

Quantity	Value
Collapse multiplier	0.144333, λ of the horizontal action along $+e_1$; hinges: intrados at x = 0.7848 m, 65.7° from the vertical; extrados at x = 4.243 m, 8.1° from the vertical; intrados at x = 8.517 m, 49.5° from the vertical; extrados at x = 10.38 m, 90° from the vertical
Mechanism through the four hinges	multiplier 0.144333 by virtual work (difference 1.2e-14)
Friction needed by the line of minimum thrust	|V| / |N| up to 0.3517
Global balance (normalised)	force 0.0e+00, moment 9.0e-17

Examples › Arch: semicircular arch of minimum thickness finds the least thickness of the semicircular arch under its own weight:

Quantity	Value
Arch	Circular arch, span 10 m, rise 5 m, radius of the centre line 5 m, thickness 1 m
Joints	201 joints normal to the centre line, 200 slices
Weight of the arch	314.159 kN (γ = 20 kN/m³, depth 1 m)
Horizontal action	none
Sliding	excluded (Heyman: no sliding)
Minimum thrust H_A	48.3936 kN, H_B = 48.3936 kN; touches intrados at x = 1.091 m, 60.3° from the vertical; extrados at x = 5 m, 0° from the vertical; intrados at x = 8.909 m, 60.3° from the vertical
Maximum thrust H_A	79.9705 kN, H_B = 79.9705 kN; touches extrados at x = -0.5 m, 90° from the vertical; intrados at x = 2.832 m, 28.8° from the vertical; intrados at x = 7.168 m, 28.8° from the vertical; extrados at x = 10.5 m, 90° from the vertical
Geometric safety factor	1.8589: the joints shrunk about their centres to 0.5379 of their length still contain a line, with extrados at x = -0.5 m, 90° from the vertical; intrados at x = 1.318 m, 54.9° from the vertical; extrados at x = 5 m, 0° from the vertical; intrados at x = 8.682 m, 54.9° from the vertical; extrados at x = 10.5 m, 90° from the vertical
Minimum thickness, loads recomputed	0.53732 of the given thickness (t = 0.537323 m at the crown), t / R = 0.10746; hinges: extrados at x = -0.2687 m, 90° from the vertical; intrados at x = 1.129 m, 54.9° from the vertical; extrados at x = 5 m, 0° from the vertical; intrados at x = 8.871 m, 54.9° from the vertical; extrados at x = 10.27 m, 90° from the vertical
Friction needed by the line of minimum thrust	|V| / |N| up to 0.3112
Global balance (normalised)	force 0.0e+00, moment 3.4e-17

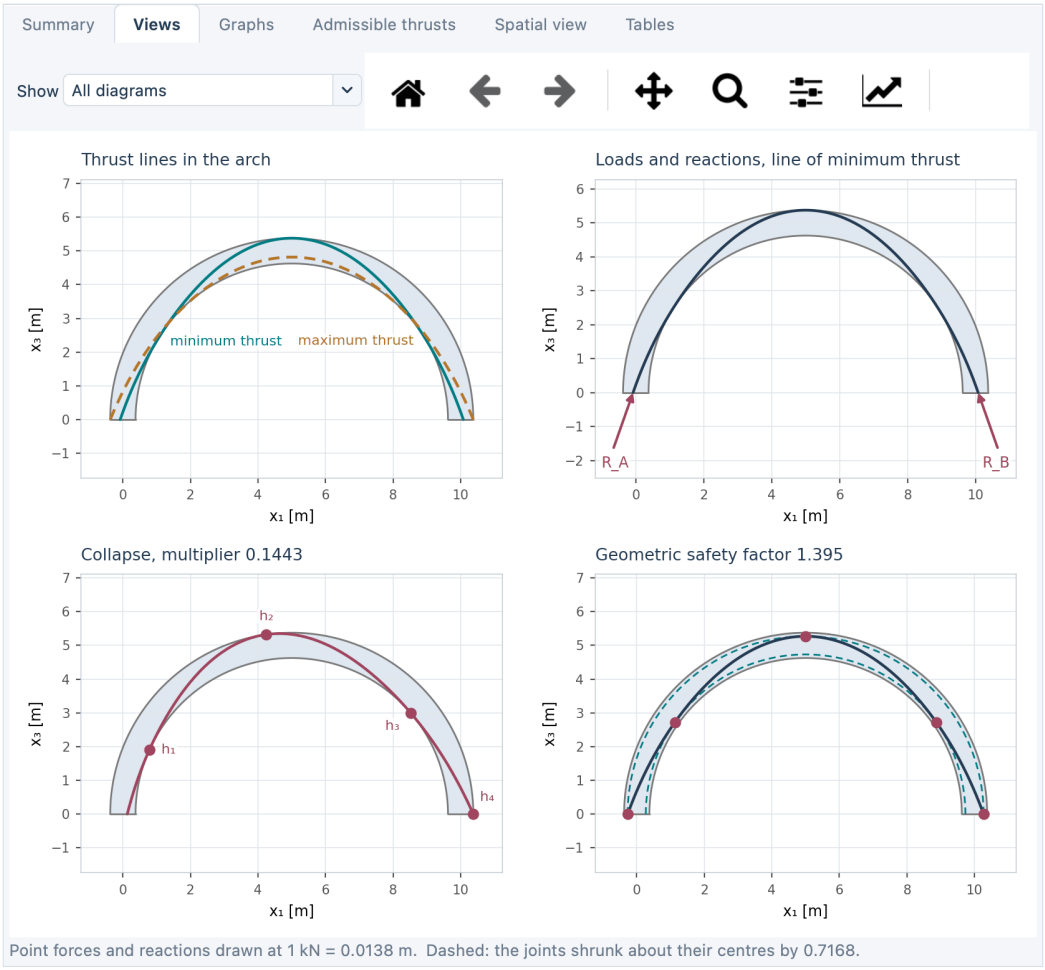


Figure 11.3. The Views tab of a semicircular arch under a horizontal action.

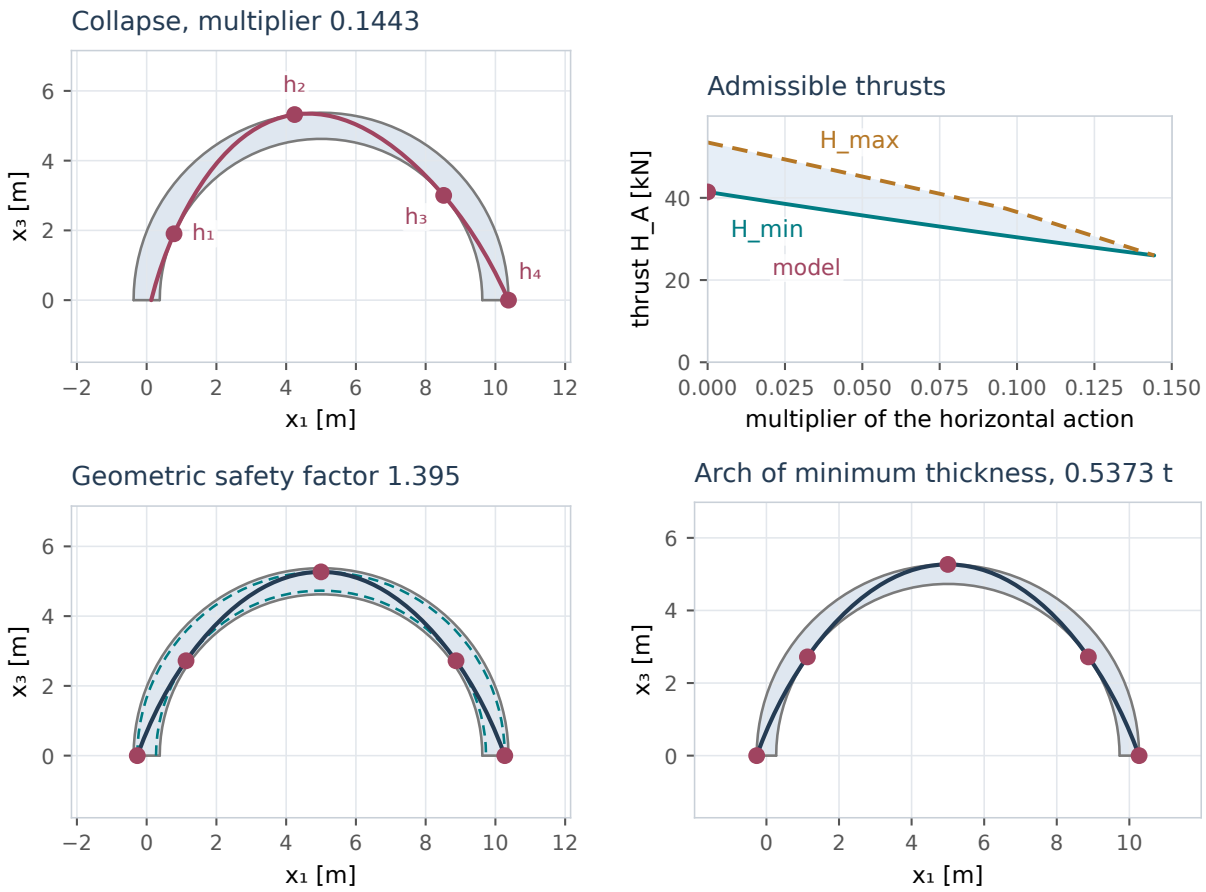


Figure 11.4. Collapse mechanism under the horizontal action, admissible thrusts against the multiplier, the reduced arch of the geometric safety factor, and the semicircular arch of minimum thickness.

12 Checks of every result

Every result carries independent checks in the Balance table:

- the resultant force and moment of all actions on the line, end forces and point actions included, which vanish for an equilibrium,
- a finite-difference test of the local equations $\mathrm{F}_{,s} + \mathrm{p} + \mathrm{r} = 0$.

The last line of the summary repeats the global balance. The values are normalised by the sum of the magnitudes of the actions. A value near the rounding error of the computation, 10^{-12} or below, confirms the computed state. A larger value either reflects the resolution, and then decreases when the samples are increased, or belongs to data that the program reports as admitting no equilibrium, such as a path that is not an axial equilibrium of its loads, whose best fit leaves a moment unbalanced. For an arch the balance is that of the selected line, with the reactions at the centres of pressure of the springings and the loads of all the slices.

The test suite of the repository checks the solvers against the closed forms and against the values reported in the article, the minimum thickness of the semicircular arch against the classical value, the examples, the files and the exports, and the clearance of the labels of every drawing.

13 Exporting results

Every export needs a solved model. The commands are in **File › Export** and in the Export button of the tool bar.

13.1 Drawings

Drawings (PDF, SVG, PNG)… opens the dialog of Fig. 13.1. Select the drawings, the format and the size:

Fit to a width. Every drawing fills the given width, from 5 to 100 cm, with its axes and grid. The margins follow the texts of the drawing: a long title or axis label is broken into lines, and the height of the sheet follows the proportions of the drawing, so that no text is cut.

Views to scale, without axes. Plans, elevations and diagrams are drawn to scale, as in a plate, with a scale bar and the title: the Length scale in cm/m applies to plans and elevations, the Force scale in cm/kN to diagrams in kN or kN/m. The scale bar is about 5 cm long, shorter under a small drawing, and the sheet widens when the title or the text of the scale bar needs it. Graphs along the line are always fitted to the width.

Resolution. The dots per inch of a PNG image.

Each drawing is written as `<base>_<diagram>.<format>`, where `<diagram>` is the key of Table 13.1. The labels are placed for the size of the file, clear of the geometry.

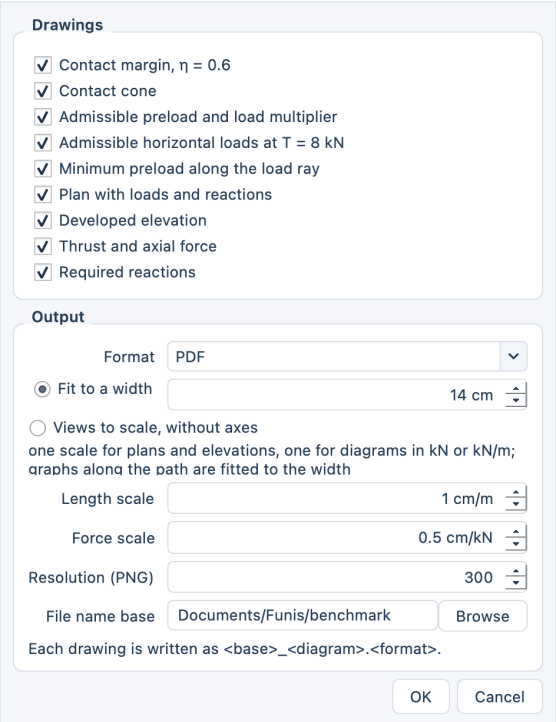


Figure 13.1. The dialog Export drawings.

13.2 DXF

Table 13.1. Keys of the drawings in the exported file names.

Key	Drawing
`plan, elevation`	Plan with loads and reactions (or of the curve), developed elevation
`forces, reactions, loads`	Thrust and axial force, required reactions, loads
`residual`	Transverse force of the best fit (path without reactions)
`curvature, plane`	Curvature and torsion, distance from the plane of C and p(0)
`newton, iterates`	Endpoint correction, elevations of the successive estimates
`margin, cone, region`	Contact margin, contact cone, admissible preload and load multiplier
`domain, preload`	Admissible horizontal loads, minimum preload along the load ray
`node_plan, node_elevation, polygon`	Node in plan and in elevation, force polygon
`funicular`	Plane funicular
`arch_lines, arch_loads`	Thrust lines in the arch, loads and reactions
`arch_collapse, arch_gsf, arch_thickness`	Collapse, geometric safety factor, arch of minimum thickness
`eccentricity, arch_forces, thrust_domain`	Eccentricity, normal force at the joints, admissible thrusts

Drawings as DXF… writes one DXF file (AutoCAD 2010 format) with every view to scale, side by side from left to right, each with its title. Plans and elevations are in metres, one drawing unit per metre. Diagrams in kN or kN/m are drawn at 0.1 drawing units per unit. Every view has its own layers, named after the drawing and the role of the line (loads, reactions and so on), and the spatial path of a spatial task is written as a 3D polyline on the layer `spatial_path`. Graphs of functions are not written.

13.3 Tables

CSV. One file with every table, each introduced by its title and followed by its note, in UTF-8.

Excel. One workbook with a sheet per table, the summary first, a styled header and frozen headings.

JSON. The title, the input of the project, the summary, every table with its columns, rows and note, the messages and, for a spatial path, the coordinates of its nodes.

13.4 Report

Report (HTML)… writes one self-contained HTML file, with the images embedded: the name and task, the notes and warnings, the summary, every drawing, every table and the complete input. It opens in any web browser and can be printed to PDF from there.

13.5 Project files

A `.funis` file is a JSON text with the keys `application`, `version` (the version of Funis that saved it), `saved` (date and time) and `project` (the input). Fields missing from an older file take their default values and unknown fields are ignored, so a project saved by one version opens in the next. Being a text file, a project can be compared, versioned and generated by a script.

14 Examples

The **Examples** menu holds 33 projects (Table 14.1): the cases of the article, with figure and section numbers referring to it, three reference cases of the application (a catenary, a parabolic arch and a free curve with a point force), and the plane cases, funiculars checked against their closed forms and arches of several shapes. The descriptions are the notes that each example shows in the Summary tab.

Table 14.1. The examples, grouped as in the menu.

Example	Description
Supported path	
Fig. 2, reaction shift on the circular path	Circular helix R = 2 m, c = −1 m/rad, $0 \le \theta \le 2.7$ rad, $p_3 = -1$ kN/m, $S_0 = -8$ kN and h = 2 kN/m along e_1. The horizontal load leaves the thrust unchanged and shifts the required reaction by $-h\ e_1$ (dashed: the reaction of the vertical load alone).
Figs. 11 and 12, circular benchmark with contact	Complete turn of the benchmark, $S = -8 - 4\theta$, r_t = 2 + h sin θ, r_n = 4 + 2θ + h cos θ. Contact with $\mu = 0.6$ and a centred strip L_sw = 0.8 m at $\delta = 0.5$ m, so $\varphi = 0.8$ and $\eta = 0.6$. The Contact tab shows the admissible loads at T = 8 kN (Fig. 12a) and the minimum preload along three directions (Fig. 12b).
Fig. 11, contact violated at h/w = 4	The benchmark with h = 4 kN/m: the line stays in compression, while the required reaction leaves the contact cone near the start, so the contact margin is negative there.
Section 7.3, pseudo-static action	Horizontal action proportional to the vertical load, $h = \gamma$ w along e_1 with $\gamma = 0.5$. The axial force is unaffected and the minimum preload is 5.67 kN, against 6.67 kN without horizontal action.
Section 7.4, point force on the helix	Point force G = (2, 0, −1) kN at θ = 5.7 rad (s = 11.4 m) with h = 1 kN/m. The thrust jumps by $-G_3$ / f,s = −2 kN and the horizontal point reaction R_h = (G_3 / f,s) t − G_h ≈ (−0.90, 1.67, 0) kN satisfies the point contact cone with $\eta = 0.6$.
Fig. 9, ellipse with f = cθ	Ellipse with semi-axes a = 3 m and b = 2 m, f = cθ with c = −1 m/rad, $p_3 = -1$ kN/m, h = 1 kN/m along e_1 and $S_0 = -8$ kN. Both elevations share the projected curvature ab/g^3 but not the grade, the axial force or the tangential reaction.
Fig. 9, ellipse with f = k_s s	Ellipse with semi-axes a = 3 m and b = 2 m, f = k_s s with k_s = -2π / L = -0.396030, for the same total drop of 2π m, $p_3 = -1$ kN/m, h = 1 kN/m along e_1 and $S_0 = -8$ kN. Both elevations share the projected curvature ab/g^3 but not the grade, the axial force or the tangential reaction.
Path without reactions	
Fig. 3, path that is not an axial equilibrium	The funicular of p = (0.7, 0, −1) kN/m and C = (−3, −4, −2) kN with 0.6 sin(πs/5) m added to its elevation. No initial force makes F = C − P parallel to its tangent.
Fig. 3, the funicular of the same loads	The funicular of p = (0.7, 0, −1) kN/m and C = (−3, −4, −2) kN, given as a table of points: the best initial force recovers C and the transverse force vanishes within the interpolation of the points.
Fig. 4, three-component loading	The thrust $S = -8 - 4\theta = -8 - 2s$ kN of the benchmark carried without reactions: the loads p = −(S u),s have three components, $p_1 = -2 \sin\theta + (-4 - 2\theta)\cos\theta$, $p_2 = 2\cos\theta + (-4 - 2\theta)\sin\theta$ and $p_3 = -1$ kN/m, and vary in direction.
Fig. 8, exceptional helical arc	R = 2 m, c = −1 m/rad, B = 3 kN, $\pi + 0.15 \le \theta \le \pi + 1.2$ rad. Loads in the plane of e_1 and e_3 conserve $F_2 = B$, so S = B / t_2 = B sec θ (compression for B > 0 inside $\pi/2 < \theta < 3\pi/2$), with h = B a / (b g $\cos^2\theta$) and $p_3 = -B\ c \sin\theta$ / (b g $\cos^2\theta$).

Example	Description
Unknown geometry (form-finding)	
Fig. 10, case A	p = (0.7, 0, −1) kN/m, C = (−3, −4, −2) kN, L = 5 m. Parallel loads: the curve lies in the plane of C and the load direction and has zero torsion. The closed form of Appendix B checks the quadrature.
Fig. 10, case B	p = (0.7 + 0.2 s, 0, −1) kN/m, C = (−3, −4, −2) kN, L = 5 m. The load direction varies and the curve leaves the plane of case A, up to about 0.32 m.
Figs. 5 and 10, case C	p = (0.7 + 0.2 s, 0.3 s, −1) kN/m, C = (−3, −4, −2) kN, L = 5 m. The final support x*(5 m) ≈ (3.58655144, 3.46958475, −0.09396228) m is an output of the construction.
Fig. 6, prescribed second support	The second support of case C is prescribed and the initial force is found by the damped Newton correction from (−2.5, −3.5, −1.5) kN: four updates reduce the mismatch from 0.37 m to below 10^{-13} m and recover (−3, −4, −2) kN.
case C with a point force	Case C with G = (0.5, 0, −1) kN at s = 2.5 m and no point reaction: the force jumps by −G and the tangent turns at the node, where the curve has a kink.
catenary under self-weight	Weight w = 1 kN per metre of actual length and horizontal thrust 4 kN: the inverted catenary $x_3 = a\,[\cosh(L/2a) - \cosh((s - L/2)/a)]$, a = H/w = 4 m, from the force ODE.
parabolic arch	w = 1 kN per projected metre and horizontal thrust 4 kN: the parabola $x_3 = w\,s\,(L - s) / (2H)$ with constant thrust S = −4 kN.
Node under a concentrated force	
Fig. 7, concentrated multidirectional force	F^- = (−3, −2, −1) kN and G = (2, 0, −1) kN give F^+ = (−5, −2, 0) kN and a change of tangent of 19.4463°. No point reaction is used.
Section 7.4, horizontal point force	F^- = (−3, −2, −1) kN and G = (2, 0, 0) kN: the tangent turns by 12.6044°.
Plane funicular	
catenary through a point	Weight w = 1 kN per metre of actual length, span 10 m, through the crown (5, 3) m: the inverted catenary z = z_c − a cosh((x − x_c)/a), a = H/w, against the integration of the force.
parabola through a point	Load w = 1 kN per metre of x, span 10 m, through the crown (5, 3) m: the parabola with constant thrust $H = w\,L^2 / (8\,f)$.
constant vertical and horizontal loads	w = 1 kN/m and h = 0.3 kN/m per metre of x: the thrust varies linearly and the line is the logarithmic branch of Appendix B (B = 0).
catenary under a pseudo-static action	Weight w = 1 kN per metre of actual length with a horizontal action 0.2 w in the same measure: the load keeps a fixed direction and the line is a catenary with an inclined axis.
funicular polygon of point loads	Three point loads and no distributed load: the funicular polygon through the crown point, with the force constant between the nodes and jumping by −G at each.
catenary with a prescribed thrust and unequal supports	w = 1 kN per metre of actual length, H = 6 kN at A and a second support 1.5 m higher: the vertical force at A that reaches B.
Arch assessment	
semicircular arch of minimum thickness	Semicircular arch R = 5 m, t = 1 m, under its own weight with joints normal to the centre line. The least thickness that contains a thrust line, the weight following the thickness, is t/R = 0.1075 (Milankovitch), with five hinges, at the springings, near 54.5° from the crown and at the crown.
semicircular arch under a horizontal action	Semicircular arch R = 5 m, t = 0.75 m (t/R = 0.15): the largest horizontal action λ W that a thrust line can carry, the four hinges of the mechanism and the same multiplier by virtual work, and the thrusts admissible at every λ.

Example	Description
segmental arch with fill and a point load	Segmental arch, span 10 m and rise 2.5 m, t = 0.6 m, fill up to 3.6 m and a point load of 50 kN at a quarter of the span: the minimum and maximum thrust and the collapse multiplier of the point load.
pointed arch under a horizontal action	Pointed arch, span 6 m and rise 4.5 m, t = 0.5 m, with a horizontal action $\lambda = 0.1$ of the weight: the lines of minimum and maximum thrust at that action and the collapse multiplier.
catenary arch of finite thickness	Catenary centre line, span 10 m and rise 4 m, t = 0.4 m, under its own weight: the line through the centres of the springings and of the crown is close to the centre line but not on it, since the weight of a ring of finite thickness acts at the centroids of its slices.
thrust line on vertical sections	The segmental arch with fill and a point load, assessed on vertical sections, the sections at constant x of the projected abscissa, in place of joints normal to the centre line.
voussoir arch of 15 voussoirs, haunched	Semicircular arch of 15 voussoirs, 0.6 m thick at the crown and 0.9 m at the springings, with the joints drawn: the lines of minimum and maximum thrust and the geometric safety factor.

15 Messages and troubleshooting

Funis distinguishes three kinds of message. An *input error* means that a field cannot be read. A message *the task could not be solved* means that the data do not admit a solution of the task, and gives the reason. A *warning*, shown in colour above the summary, accompanies a solution that holds but deserves attention. Table 15.1 lists the most frequent ones.

Table 15.1. Frequent messages and what to do.

Message	Meaning and remedy
Row …, column …: '…' is not a number	A cell of a table holds text. Correct the cell.
The helix needs $R > 0$ and an end angle beyond the start angle	Check the radius and the angles.
The plan of the path stops or reverses	Two points of the table have the same plan position, or the plan turns back. Reorder or thin the points.
The path has a stationary elevation	A supported path needs a non-zero grade, since the thrust is the vertical force divided by the grade. Give the path a grade everywhere.
The line is not in compression	The thrust becomes positive somewhere. Raise the preload or check the loads.
The required reaction leaves the contact cone	The wall cannot supply the reaction with the given friction and strip. Read the admissible preloads and multipliers in the summary and in the Contact tab.
The contact cone needs $\mu > 0$ and a positive strip width	Correct the contact data, or switch off the contact check.
The assigned path is not compatible with these loads without reactions	No initial force makes the path an axial equilibrium. Use a supported path, or find the curve by form-finding.
The horizontal force vanishes	The curve would turn vertical, where the projected length stops being a parameter. Change the initial force or the loads.
The endpoint correction stopped before reaching the target	The second support may be out of reach. Improve the estimate of C or check the target.
The horizontal force F_1 is not negative	The plane line would turn vertical or into tension. Raise the thrust or change the point of the line.
The point of the line must lie between the supports	Use $0 < x_c < L$.
No thrust line lies within the arch under these loads	By the safe theorem the arch is not shown to be safe. Read the geometric safety factor, below 1, and the eccentricities.
The line through the three points lies outside the arch	The prescribed line is not admissible at the listed joints. Other lines may be.
The … folds back: the thickness exceeds twice the radius of curvature	The intrados of a strongly curved centre line crosses itself. Reduce the thickness or change the shape.
A pointed arch needs a rise larger than half the span	Raise the rise, or use a circular arch.
The point load at $x =$ … is not above the extrados of the arch	Move the load within the span of the extrados.
Unexpected error	An error of the program. Save the project and report it with the file on the issue page of the repository.

16 Hypotheses and limitations

- Axial equilibrium. Every task computes equilibrium states of lines carrying an axial force alone. It does not compute deformations, stresses in a section, buckling or dynamic response. A horizontal action is a pseudo-static load.
- Spatial paths. The projected length must be a regular parameter: the plan cannot stop, turn back or become vertical. A supported path needs a non-zero grade everywhere. The lateral support supplies horizontal reactions only.
- Contact. The cone describes the wall as a support with friction and, optionally, a centred strip. A violated cone is a statement on the support, not on the line, and the admissible intervals are not collapse loads.
- Form-finding. The horizontal force must stay positive. A second support is reached by a local correction from the estimate, which may fail for a target far from it.
- Plane funicular. The line must remain a function of x, with $F_1 < 0$ everywhere.
- Arches. The hypotheses of Heyman hold: no tensile strength, unlimited compressive strength and, with a zero friction coefficient, no sliding. The fill loads the arch but has no strength and no passive resistance. The arch is cut into a finite number of slices, so the results converge with the number of slices.

A Quick reference

A.1 Keyboard shortcuts

Command	Windows	macOS
New project	Ctrl+N	⌘N
Open	Ctrl+O	⌘O
Save	Ctrl+S	⌘S
Save as	Ctrl+Shift+S	⇧⌘S
Run	F5	F5
User guide	F1	⌘?
Quit	Alt+F4	⌘Q

A.2 Command line

Command	Effect
`Funis project.funis`	Open the window with a project.
`Funis --selftest [image.png]`	Solve every example without showing the window, save a screenshot and a report `image.log`, and exit with 0 when every example solves (Section 2.5).
`python run_funis.py [project.funis]`	The same from the source code.
`python -m funis`	Start from the source code.

A.3 Notation

Symbol	Meaning
$\hat{e}_1, \hat{e}_2, \hat{e}_3$	Unit vectors of the axes, $\hat{e}_3$ upward
s, L	Projected arc length and projected length of a spatial path
$x^*(s)$, $x(s)$, $f(s)$	Spatial path, its plan and its elevation
$\hat{t}$, $\hat{n}$, ρ	Projected unit tangent, left normal and projected curvature
$f_{,s}$, J	Grade and ratio of actual to projected length
u, $\hat{T}$	Tangent vector $\hat{t} + f_{,s}\hat{e}_3$ and unit tangent of the spatial path
F, S, N, V	Force, thrust, axial force and vertical force
C, P	Force at the start and primitive of the loads
p, r	Load and reaction per projected metre
w, h, β, γ, λ	Vertical and horizontal load densities, direction of the horizontal load, pseudo-static coefficient, multiplier of the horizontal load
T	Preload, $T = -S(0)$
r_t, r_n, r_ν	Tangential, normal and wall-normal reaction
μ, L_{sw}, δ, φ, η	Friction coefficient, strip width, distance of the strip from the wall, strip limit and effective coefficient of the cone
G, R	Point force and point reaction
Q	Horizontal force of a free curve
κ, τ	Curvature and torsion of a free curve
B, $\hat{m}$, $\hat{d}$	Conserved component, normal and direction of a vertical load plane

Symbol	Meaning
x, z, H	Abscissa, height and thrust in the plane tasks
t, b, e	Thickness and depth of an arch, eccentricity of the centre of pressure

References

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[3] J. Heyman, The stone skeleton, *International Journal of Solids and Structures* 2 (1966) 249–279, doi:10.1016/0020-7683(66)90018-7.

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